



Discover the Experience

EB tresos[®] Studio

release notes

product release 14.2



EB tresos® Studio

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1. Release notes EB tresos Studio

1.1. Version information

Release notes for EB tresos Studio 14.2.

1.2. Interpretation of version information

The EB tresos *product line release* schedule contains two product line releases per year. Product line releases are named after the year of release and a variant represented by a letter. Examples for product line release version numbers are *2009.a*, *2009.b*, *2010.a*, etc.

The EB tresos product line version number can be found on the splash screen, the commandline and the *EB tresos details* dialog.

EB tresos Studio is released as part of the EB tresos product line. Each product within the product line is assigned an individual *product version number*.

The EB tresos Studio product version number can be found on the splash screen and the *EB tresos details* dialog next to the keyword *Studio*.

The EB tresos Studio product version number scheme is made of three parts: the major, the minor and the patch number. The numbers are separated by dots in the following naming scheme: `<major>.<minor>.<patch>`.

- ▶ An increment of the *major* version number indicates that the release contains major new features and changes compared to the previous major version.
- ▶ An increment of the *minor* version number indicates that the release contains minor new features and changes compared to the previous minor version.
- ▶ An increment of the *patch* version number indicates that the release fixes known problems.

There may be a *qualifier* additionally to the product version number. This *qualifier* is provided in brackets and displayed after the product version number. Examples for qualifiers are: *alpha*, *beta*, *customer-exclusive pre-release*, ...

- ▶ If the qualifier is *alpha* (e.g. *10.0.3 (alpha)*), the release is a pre-release. Pre-release means that not all planned features are implemented (i.e. not *feature complete*) and the pre-release just fulfills basic quality tests.
- ▶ If the qualifier is *beta* (e.g. *10.0.4 (beta)*), the release is a pre-release that implements all planned features but only fulfills basic quality tests.

Examples for spelled out EB tresos Studio product versions are:

- ▶ *EB tresos Studio 10.0.0 (alpha)*
- ▶ *EB tresos Studio 10.0.4 (beta)*
- ▶ *EB tresos Studio 10.0.6*
- ▶ *EB tresos Studio 10.1.0 (exclusive for customer xyz)*

1.3. Limitations and restrictions

Summary:	SWTException with guided configuration wizards when using browser widgets inside expandable composites (Win7 only)
Description:	The problem only occurs on computers with OS Win7. An SWTException appears when running unattended wizards via the unattended wizard configuration dialog or when closing a Sidebar dialog (Finish/Cancel). The problem only occurs for wizards showing a browser widget located inside an expandable group section.
Rationale:	This is due to a bug inside eclipse SWT https://bugs.eclipse.org/bugs/show_bug.cgi?id=412706 .
Workaround:	If you are developing a guided configuration wizard and want to use a browser widget, then use for all parent groups <code>WidgetFactory.GroupDecoration.LINE</code> instead of <code>WidgetFactory.GroupDecoration.TITLE</code> .
ID:	TRESOS-3569

Summary:	Exported AUTOSAR system description files cannot always be validated against their schema files
Description:	If the EB tresos Studio system model is exported into an AUTOSAR system description file, it cannot be guaranteed that the file can be validated against its AUTOSAR schema file.
Rationale:	Non AUTOSAR file formats, such as DBC, LDF, and Fibex do not provide all the data that AUTOSAR system models require. The importers for these file formats do not configure unavailable data in the EB tresos Studio internal system model, leaving it "incomplete". If such "incomplete" EB tresos Studio system models are exported to AUTOSAR system description files, these files won't validate against their AUTOSAR schema file due to the missing data. Similar validation problems will arise if an AUTOSAR system description file of an older revision, e.g. AUTOSAR 3.1 is exported into an AUTOSAR system description file of a more recent revision, e.g. AUTOSAR 4.0.
ID:	TRESOS-3495

Summary:	The VSMD rule EcucSws_6046 is not always checked
Description:	The requirement EcucSws_6046 defines that the <code>DESTINATION-REF</code> of a reference shall start with <code>/AUTOSAR/EcucDefs</code> , if it points to a StMD. This is only checked by the command legacy vsmdcheck if the references point into a VSMD that is contained within one of the input files. Otherwise no violation is created.
Rationale:	EB tresos Studio does not support to define a file containing all available StMDs, so it is not possible to check this rule fully.
ID:	TRESOSWP-2050

Summary:	The preference <i>General/Always run in background</i> cannot be disabled in EB tresos Studio.
Description:	You cannot change the value of the Eclipse preference <i>General/Always run in background</i> . It is always forced to <i>true</i> by EB tresos Studio.
Rationale:	Setting this option to <i>false</i> would lead to the consumption of unnecessary resources in the operating system and eventually to a crash of EB tresos Studio

Summary:	The <code>ADMIN-DATA</code> tag is not supported in files prior to AUTOSAR 3.0.
Description:	EB tresos Studio ignores the <code>ADMIN-DATA</code> tag and all sub-tags in AUTOSAR files prior to AUTOSAR 3.0.
Rationale:	Support for files prior to AUTOSAR 3.0 has not been implemented.

Summary:	<code>BSWModuleTemplate</code> is no longer imported
Description:	EB tresos Studio no longer imports or exports the ARElement <code>BSWModuleTemplate</code> for AUTOSAR 3.x releases.
Rationale:	The information from the <code>BSWModuleTemplate</code> is not needed to configure BSW modules prior to AUTOSAR 4.0.

Summary:	The value of the <code>DEST</code> attribute is ignored
Description:	The value of the <code>DEST</code> attribute is ignored. If the appropriate reference-target does not match the type specified by the <code>DEST</code> attribute, no error or warning is created.
Rationale:	EB tresos Studio does not need the value of the <code>DEST</code> attribute to follow references. Support for the <code>DEST</code> attribute would waste memory.

Summary:	When you convert or merge files, the <code>DEST</code> attribute requires an appropriate reference-target
Description:	When converting or merging files, the <code>DEST</code> attribute for <code>REFERENCE</code> values and <code>CONFIGURATION-CLASS-AFFECTION</code> is only written correctly, if the appropriate refer-

	ence-target exists. If the target does not exist, a default-value is written to ensure validity of the converted or merged XML file.
Rationale:	As EB tresos Studio does not store the value of the <code>DEST</code> attribute the value can only be determined from the reference-target.

Summary:	If a configuration with missing values is converted or exported to an AUTOSAR file format, the XML tag <code>VALUE</code> is suppressed.
Description:	If a configuration lacks values and is converted or exported to an AUTOSAR file format, the XML tag <code>VALUE</code> is suppressed. Even though the file validates against the AUTOSAR XSD file, suppressing this tag is not allowed according to the <i>Specification of ECU Configuration</i> .
Rationale:	If the configuration lacks a value a valid value cannot be determined.
ID:	TRESOSWP-1063

Summary:	It is not possible to work with sub-sets of parameters
Description:	To avoid editing the wrong kind of parameters, it is not possible to work with sub-sets of parameters. Sub-sets of parameters can be e.g. pre-compile time parameters.
Rationale:	This feature has not been implemented.
ID:	[ecuc_sws_4003]

Summary:	Only 64-bit signed integers are supported
Description:	Currently, only 64-bit signed integers are supported. Greater values such as 64-bit unsigned integers cannot be used.
Rationale:	EB tresos Studio uses native Java data types for storing values to get a better performance and a better memory footprint. The largest native integer format provided by Java is 64bit signed integer.

Summary:	EB tresos Studio can handle numbers only up to 56 bit in XPath expressions.
Description:	The XPath engine used in EB tresos Studio can handle numbers only up to 56 bit. However, the functions in the namespace <code>num:</code> are able to deal with 64-bit signed numbers if the input is provided as string.
Rationale:	This is a limitation introduced by the XPath engine, on which EB tresos Studio is built.

Summary:	The XPath function <code>num:order()</code> removes duplicate elements
Description:	The XPath function <code>num:order()</code> removes duplicate elements. This behavior is intended.
Rationale:	Although you may expect this function to be implemented in a different way, the EB tresos Studio implementation meets the XPath specification.

Summary:	The EB tresos Studio GUI fails to start when it is installed under the root directory
Description:	If you have installed EB tresos Studio directly under the root directory, or if a drive letter is assigned to the EB tresos Studio base directory (e.g. X:\bin\tresos_gui.exe), you cannot start the EB tresos Studio GUI. In this case an error message appears saying that the Java Virtual Machine has been terminated. The EB tresos Studio command line applications are not affected by this issue. Two alternative workarounds are available: 1. To start the EB tresos Studio GUI with EB tresos Studio installed under the root directory, specify the EB tresos Studio base directory in the file <code>bin\tresos_gui.ini</code> with the EB tresos Studio command line: ▶ Right before line <code>-vm</code>, enter the line <code>-install</code>. ▶ After the line <code>-install</code>, enter a line containing your EB tresos Studio base directory and terminate the EB tresos Studio base directory with a slash. E.g.: <pre>-install X: /</pre> 2. If you always start <code>tresos_gui.exe</code> from the <code>bin</code> directory, specify <code>..</code> as EB tresos Studio base directory.
Rationale:	This problem occurs, because the underlying Eclipse startup code that searches for the current installation location is inaccurate.
ID:	TRESOS-2364

Summary:	Importing an AUTOSAR 4.0 system description file may fail
Description:	There are several limitations to the System Description Importer for AUTOSAR 4.0 files: ▶ The AUTOSAR documentation feature is not supported and is silently ignored when you import a file. ▶ The formula language feature defined by AUTOSAR is not supported. Therefore, if you import a file that contains the formula language, the import fails and an error message opens up.
Rationale:	

Summary:	Signal based gateway information cannot be imported from the AUTOSAR 3.1.4 format
Description:	EB tresos Studio does not support <code>SignalMapping</code> elements in AUTOSAR version 3.-1.4 system description files. If you import an AUTOSAR version 3.1.4 system description

	file that contains <code>SignalMapping</code> elements into EB tresos Studio, these <code>SignalMapping</code> elements are skipped and a warning is reported.
Rationale:	In the system description files of AUTOSAR version 3.1.4, a major change was done to <code>SignalMapping</code> elements. Therefore, AUTOSAR version 3.1.4 <code>SignalMapping</code> elements are incompatible to older versions.
Summary:	System Description Importer: AUTOSAR version 3.1.4 <code>CanCommunicationController</code> elements are only partly supported
Description:	EB tresos Studio only supports parts of the AUTOSAR version 3.1.4 system description file elements <code>CanCommunicationController</code> : The subelement <code>CanCommunicationControllerAttributes/CanControllerConfigurationRequirements</code> and the subelement attribute <code>CanCommunicationControllerAttributes/CanControllerConfiguration/propagationDelay</code> are not supported. If you import an AUTOSAR version 3.1.4 system description file that contains one or more of the named elements into EB tresos Studio, these elements are skipped and a warning is reported.
Rationale:	In the system description files of AUTOSAR version 3.1.4, a major change was done to <code>CanCommunicationController</code> elements. Therefore, AUTOSAR version 3.1.4 <code>CanCommunicationController</code> elements are incompatible to older versions.
Summary:	Default buffer assignments may lead to high CPU load and to manually created hardware objects not receiving any CAN-IDs anymore
Description:	If you configure the modules <code>Can</code> and <code>CanIf</code> in a configuration import and select Create default buffer assignment in the Buffer Assignment in Can/CanIf drop-down list box, up to two CAN Rx and up to two Tx hardware objects are created. These default objects handle all standard and extended CAN-IDs your ECU is sending and receiving. The Rx hardware objects have their hardware filters turned off, which means the <code>CanIf</code> must filter for valid incoming CAN-IDs in software. This may lead to a high CPU load if your ECU receives many PDUs. Moreover, hardware objects that were created manually before the configuration import will no longer receive any CAN-IDs after the configuration import.
Rationale:	The configuration importer cannot make any assumptions on the properties of the CAN cells of your ECU and therefore creates this default assignment which works on virtually every CAN cell. In order to obtain a buffer assignment tailored to the CAN cells of your ECU, use the <i>CAN Buffer Assignment Editor</i> , which is described in the EB tresos AutoCore com stack users guide.
Summary:	During import, large integer values may be rounded for attributes of AUTOSAR type <i>Numerical</i>
Description:	EB tresos Studio can only handle values which can be represented as double precision float value according to <i>IEEE 754-1985</i> for all AUTOSAR attributes of type <code>Numerical</code> . Values which cannot exactly be represented in this format will be rounded during import to the nearest representable value and an appropriate warning will be reported.

Rationale:	Numerical values may also hold floating point values and are internally stored with use of the native Java type <code>double</code> for performance reasons. The Java <code>double</code> type only supports the named precision.
ID:	TRESOS-3306

Summary:	The XPath engine evaluates calculations with infinity produce a non IEEE754 compliant result
Description:	If you get an infinite value as an intermediate result of a calculation in the XPath engine, further calculations with this value yield a result that is not conforming to the IEEE754 standard. Infinite values cannot be used directly in calculations, but you can get them as intermediate results, e.g. by computing <code>1 div 0</code>. Examples: ▶ <code>[!(1 div 0) mod 1.0"!] returns 0.0 but should be NaN</code> ▶ <code>[!(1 div 0) mod 2.0"!] returns 1.0 but should be NaN</code> The affected functions are: ▶ the XPath function <code>mod</code> ▶ public API Custom XPath functions using methods <code>AbstractXPathFunctions.toInt</code> or <code>AbstractXPathFunctions.toInts</code> ▶ any public API Custom XPath functions with a long as parameter
Rationale:	The XPath engine used in EB tresos Studio usually represents floating point numbers with the Java data type <code>double</code> . But for some calculations they are converted to integer numbers which are represented by the Java data type <code>long</code> . During this conversion, the "infinity" value is lost and accidentally replaced by the largest representable long value, which is <code>9223372036854775807</code> . This value is then used for the following calculations and thus yields in wrong results.
ID:	TRESOS-3272

Summary:	In 32 bit mode you can only import Fibex files < ~22 MB
Description:	Larger files will yield an error during ECU retrieval in the import dialog.
Rationale:	The 32 bit mode restricts the amount of available heap space which prevents the XML parser to operate properly.
Workaround:	In 64 bit mode, you can import considerably larger Fibex files. On hosts that only support the 32 bit mode, you can convert files not much larger to AUTOSAR system description files when you use <code>tresos_cmd.bat/legacy convert</code> .

Summary:	Fibex importer always uses first conversion formula for configuring the upper and lower limits of a signal
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Description:	Fibex files offer the possibility to define more than one conversion formula for converting a signal value from its internal representation (representing the signal bistream on the bus) to its physical one (representing the actual physical value of the signal). These multiple functions are to be applied on different, non overlapping intervals of the signal value range. The Fibex specification refers to this as SCALE-LINEAR conversion. Since Fibex is specifying the upper and lower limits of a signal in their physical representation but AUTOSAR requires the internal one, the Fibex importer must apply the conversion itself. It however, only uses the first available conversion formula, even if two or more formulas are provided. This may result in incorrectly imported upper and lower limits of a signal.
Rationale:	As for now, SCALE-LINEAR conversion functions have not been found in production Fibex files.
ID:	TRESOS-3269

1.4. Known problems

Summary:	AUTOSAR system description files written by EB tresos Studio always use Windows line endings
Description:	AUTOSAR system description files that are written by EB tresos Studio, e.g. via the <i>System Description Exporter</i> , always use Windows line endings (<code>\r\n</code>). That is independent of which setting you choose for the option <i>Use Unix line endings for configuration data files</i> .
Workaround:	You can convert the file to use Unix line endings (<code>\n</code>) with the use of an external tool or editor, e.g. the free tool <code>dos2unix</code> .
ID:	TRESOS-2843

Summary:	If a function processes the result of an inner function, XPath's compute the wrong value
Description:	If a function processes the result of an inner function, XPath's compute the wrong value under certain conditions. The conditions are: ▶ The result of the inner function is a list, which is filtered by a predicate expression. ▶ Additionally the outer function is able to handle lists and single values at the same time. If all these conditions apply, the result of the inner function is always the last element of the list result, instead of the correct result.

	An example of an XPath expression which computes the wrong result is e.g. <code>num:i(text split('1,2,3,4,5',' ')[position() = 2])</code> . This expression returns '5' instead of '2'.
Workaround:	No workaround is available.
ID:	TRESOS-1886

Summary:	You can only correctly import ECU configuration projects if the configuration XDM files are located beneath the project path of the imported project
Description:	You can only correctly import ECU configuration projects if the configuration XDM files are located beneath the project path of the imported project.
Workaround:	If you are adding modules to a project, either keep the default file location or make sure to select subdirectory of the project directory.
ID:	TRESOS-1343

Summary:	The System Description Importer and the tresosDB Importer cannot detect duplicate SHORT-NAMES
Description:	The System Description Importer and the tresosDB Importer cannot detect duplicate SHORT-NAMES. As a result, duplicate SHORT-NAMES may cause you to loose configuration parameters. However, modules such as e.g. the Rte may rely on these parameters and may report problems due to the loss of parameters.
Workaround:	No workaround is available.
ID:	TRESOS-2016

Summary:	The <code>legacy convert</code> command of the EB tresos Studio commandline ignores the <code>output-path</code> option (<code>-o</code>)
Description:	If you use the command <code>legacy convert</code> on the EB tresos Studio command line to convert files, specifying an output-path with the <code>-o</code> switch has no effect.
Workaround:	Specify the output file with an absolute path or with a path relative to the current working directory.
ID:	TRESOS-2197

Summary:	The XPath engine uses not existing nodes in calculations
Description:	In XPath expressions you may use node names that do not exist or that are optional and are disabled in the current module configuration. If you use such nodes in XPath expressions that do calculations based on the value of the nodes, the XPath engine assumes values for the nodes even if the nodes do not exist. For optional nodes this only leads to unexpected results, if the node in question is disabled in the module configuration on which the expression is executed. So, e.g. for the

template-based code generator, this behavior is only observed, if the node in question is used in the code template, and if it is disabled in the current module configuration.

The value of not existing nodes, which is strictly speaking *nothing*, is converted into the data type that is required by the expression operands and by the expression type:

- ▶ If the expression is a mathematical operation, i.e. if the expression uses one of the operators: `+`, `-`, `div`, `*`, or `mod`, not existing nodes are treated as if they had the value `0.0`:
 - ▶ If for example `NotExistingNode` is a not existing node, the expression `NotExistingNode + 3.0` returns `3.0`.
- ▶ If the expression is an equality check, i.e. if the expression uses the `=` operator, the result is always `false`:
 - ▶ If, for example, `NotExistingNode` is a not existing node, the expression `NotExistingNode = 'somevalue'` returns `false`. In this case the second operand used for the comparison is not taken into account. Thus, the same node from the example before, the expressions `NotExistingNode = ''`, `NotExistingNode = true()`, `NotExistingNode = false()`, `NotExistingNode = NotExistingNode` all return `false`.
- ▶ If the expression is an inequality check, i.e. if the expression uses the `!=` operator, the result is always `true`:
 - ▶ If, for example, `NotExistingNode` is a not existing node, the expression `NotExistingNode != 'somevalue'` returns `true`. The second operand used for the comparison is not taken into account. E.g. with the same node from the example before, the expressions `NotExistingNode != NotExistingNode` also returns `true`.
- ▶ If the expression is a relational expression, i.e. if the expression uses one of the operators `<`, `<=`, `>` or `>=`, the value `0.0` is used in the calculations:
 - ▶ If, for example, `NotExistingNode` is a not existing node, the expressions `NotExistingNode < 0` and `NotExistingNode > 0` both return `false`. However, the expressions `NotExistingNode <= 0` and `NotExistingNode >= 0` both return `true`.

Although the node behaves as being equal to `0.0`, please keep in mind that the equality check is not `true`, as is explained in the second item of this list.

- ▶ If the expression is a unary expression, i.e. if the expression uses the unary operator `-`, the value `0.0` is used in the calculation. Therefore, the entire expression is evaluated as `-0.0`.
- ▶ If the expression is a union expression, i.e. if the expression uses the union operator `|`, an empty node-set is used in the calculation.

	If a not existing node is passed as a parameter to a function defined by the XPath standard, the value of the not existing node is converted into the data type that is required by the function (e.g. to <code>0.0</code>, to the empty node-set, or to <code>false</code>). Functions defined by the XPath standard are easy to recognize: either they have no namespace or the namespace is <code>fn</code>. If a node is passed as a parameter to the function supplied by EB tresos Studio as an extension of the XPath engine, an error message is shown. However, this only applies to nodes that are provided as parameter without any additional calculations. If an expression is passed as an argument, the expression is evaluated by the XPath engine.
Workaround:	Use the <code>node:empty(...)</code> or <code>node:exists(...)</code> XPath functions to explicitly test whether a node exists. As you can see in the examples above, using not existing nodes can lead to unexpected results.
ID:	TRESOS-2453

Summary:	The EB tresos Studio legacy generate command line can only handle one configuration of one module type correctly in one session
Description:	If you supply more than one configuration for one module type, e.g. Can, CanIf, Com, etc., to the legacy code generation commands legacy make, legacy generate, or legacy verify, you may receive incorrect verification results or even incorrect code. AUTOSAR allows you to supply more than one configuration for one module type for some driver modules, e.g. Wdg. In this case the command line could look as follows: <pre>tresos_cmd legacy generate Wdg1.arxml@asc3.1 Wdg2.arxml@asc3.1</pre> However, depending on which code generators the involved modules use, the code generators may access wrong configurations. This problem occurs when: ▶ code generators access the variables <code>enabledConfigs</code> or <code>generateConfigs</code>, or ▶ when the code generators reference parameters via relative paths in XPath expressions and expect that XPath expressions can be resolved relative to the <code>MODULE-CONFIGURATION</code> node, or ▶ when code generators use the XPath function <code>as:modconf()</code> and always access the first returned module configuration. ▶
Workaround:	Generate configurations in separate runs of the command line.
ID:	TRESOS-2603

Summary:	Separate imports of AUTOSAR system description files can lead to wrong warnings
Description:	If you import two system description files via two System Description Importers, and one of the imported file references elements from the other imported file, the System Descrip-

	tion Importer prompts a warning about invalid reference values after the import. This happens for actually broken references as well as for references, which are resolvable using the other file.
Workaround:	Ignore the warning. To avoid the warning, import both files with the same importer.
ID:	TRESOS-2810

Summary:	Code generation does not work properly with modules in which <code>allowMultiple</code> is set to <code>true</code>
Description:	Modules in which <code>allowMultiple</code> is set to <code>true</code> in the extension and which have more than one configuration in the project, cannot be properly generated with some code generators: ▶ Template-based generators do not work at all with multiple configurations because you cannot change the output file name. This is a design limitation and no known problem. ▶ If you disable one of the configurations, the template-based code generator and the NG generation framework might operate on the wrong configuration node. E.g. if you have two configurations <code>Config_1</code> and <code>Config_2</code> and if you disable the configuration <code>Config_1</code>, the context node of the generators actually points to <code>Config_1</code> instead of to <code>Config_2</code>. However, you can still only access the nodes of <code>Config_2</code>. ▶ If you are using external generators and if you set the option <code>generateIntoOneFile</code> to <code>false</code>, only one configuration is generated. The generated configuration is the first enabled configuration that is listed in the <code>preferences.xdm</code> file. ▶ If you use multiple configurations on the legacy mode command line, only the first configuration is generated. The behavior is similar to a project with multiple configurations for all generators.
Workaround:	
ID:	TRESOS-2816

Summary:	Calling <code>DCtxt.opt.setExistingNodesOnly(true)</code> might have no effect
Description:	Calling the method <code>DCtxt.opt.setExistingNodesOnly(true)</code> has no effect, if the context currently points to a non-existing node or to an invalid node. However, if you do the opposite and call <code>DCtxt.opt.setExistingNodesOnly(false)</code> , the option is set to <code>false</code> .
Workaround:	Always check the return value of the method and implement code that handles <code>false</code> as return value.
ID:	TRESOS-2771

Summary:	The AUTOSAR element does not support the <code>ADMIN-DATA</code> attribute
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Description:	If you import an AUTOSAR file, the <code>ADMIN-DATA</code> attribute of the <code>AUTOSAR</code> element is not imported.
Workaround:	No workaround is available.
ID:	TRESOS-2728

Summary:	The <i>Compare</i> editor for XDM files does not behave correctly when you close, rename or delete the project
Description:	If you have changed an XDM file with the <i>Compare</i> editor for XDM files and then close, rename or delete the project in which this XDM file is stored, the <i>Compare</i> editor for XDM files does not handle those changes correctly: ▶ When you close the project, the <i>Compare</i> editor for XDM files does not close, too. ▶ After you have renamed or deleted the project, the <i>Compare</i> editor for XDM files saves the edited file always under its former path, when you want to save the changes.
Workaround:	No workaround is available.
ID:	TRESOS-2956

Summary:	Some XPath functions are incompatible with the built-in XPath functions
Description:	If you combine a built-in XPath function with an XPath function provided by EB tresos Studio, some combinations return <code>NaN</code> instead of the correct result. This is caused by incompatibilities in the internal number representation. If you print out the values, you will not see a difference between an incompatible and a compatible number. For example, the expression <code>num:mul(2,1.1)</code> returns <code>2.2</code>. But if you pass the result to the <code>floor</code> function, the expression <code>floor(num:mul(2,1.1))</code> returns <code>NaN</code> instead of <code>2.0</code>. Furthermore, in such cases the <code>round</code> function returns <code>0.0</code>. E.g. <code>round(num:mul(2,1.1))</code>. However, not all combinations of functions are affected. The affected built-in functions that return <code>NaN</code> when combined with a EB tresos Studio function are: ▶ number, e.g. <code>number(num:mul(2,1.1))</code> ▶ floor, e.g. <code>floor(num:mul(2,1.1))</code> ▶ ceiling, e.g. <code>ceiling(num:mul(2,1.1))</code> The affected EB tresos Studio functions that lead to <code>NaN</code> when passed to built-in functions are: ▶ <code>num:add</code>, e.g. <code>floor(num:add(2,3))</code> ▶ <code>num:div</code>, e.g. <code>floor(num:div(2,3))</code>

	▶ <code>num:mod</code>, e.g. <code>floor(num:mod(5,2))</code> ▶ <code>num:mul</code>, e.g. <code>floor(num:mul(5,2))</code> ▶ <code>num:negate</code>, e.g. <code>floor(num:negate(5))</code> ▶ <code>num:sub</code>, e.g. <code>floor(num:sub(4.8,2))</code> ▶ <code>num:order</code>, e.g. <code>floor(num:order(4.8))</code> The <code>round</code> function always returns <code>0.0</code> if the argument is one of the broken EB tresos Studio functions. The following EB tresos Studio functions have been verified to work correctly: ▶ <code>num:i</code>, e.g. <code>floor(num:i(4.8))</code> ▶ <code>num:f</code>, e.g. <code>floor(num:f(4.8))</code> ▶ <code>num:min</code>, e.g. <code>floor(num:min(4.8))</code> ▶ <code>num:max</code>, e.g. <code>floor(num:max(4.8))</code> ▶ <code>num:hextoint</code>, e.g. <code>floor(num:hextoint('0x22'))</code> ▶ <code>num:octtoint</code>, e.g. <code>floor(num:octtoint('022'))</code> ▶ <code>num:bintoint</code>, e.g. <code>floor(num:bintoint('b101'))</code> ▶ <code>num:dectoint</code>, e.g. <code>floor(num:dectoint('12'))</code> ▶ <code>num:radixtoint</code>, e.g. <code>floor(num:radixtoint('b1110'))</code>
Workaround:	Use the <code>num:f</code> function to convert a number into a compatible format, e.g. <code>ceiling(num:f(num:mul(2,1.1)))</code> .
ID:	TRESOS-3020

Summary:	The <i>Compare</i> editor for XDM files does not recognize intermediate changes of the edited file
Description:	If the file currently edited in the <i>Compare</i> editor for XDM files changes, the <i>Compare</i> editor for XDM files will overwrite those changes when you save your changes in the <i>Compare</i> editor for XDM files. For example, if you update the XDM file via your version control system, or edit the XDM file in a different text editor than the <i>Compare</i> editor for XDM files while having the same file opened in your <i>Compare</i> editor for XDM, the SVN update or the edit of the text editor are being overwritten when you save the XDM file from the <i>Compare</i> editor for XDM files.
Workaround:	No workaround is available.
ID:	TRESOS-2962

Summary:	The Save button is enabled for guided configuration wizards opened from the <i>Side-bar</i> view
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Description:	Since EB tresos Studio version 13.0, the Save button is enabled by default, when you work on guided configuration wizards opened from the <i>Sidebar</i> view. However, clicking the Save button has no effect.
Workaround:	No workaround is available.
ID:	TRESOS-3407

Summary:	Expression language can only deal with defined variables
Description:	Some parts of the EB tresos Studio Public APIs allow you to define Boolean expressions via XML tags in the <code>plugin.xml</code> file. These expressions can access the values of pre-defined variables. However, all variables that are used must be defined. If you access undefined variables, the associated expression immediately evaluates to <code>false</code> as soon as the parser encounters the access to the variable. The behavior of the parser is deterministic, but it contradicts usual programming conventions. For example, the order of nested expressions is important for the result. In the following example, the variable <code>FOO</code> is set to <code>enabled</code> and the variable <code>BAR</code> is undefined. The following code evaluates to <code>true</code>. The parser first encounters the variable <code>FOO</code>. Since the variable is defined and equals the value of the nested <code>equals</code> statement, the <code>or</code> expression evaluates to <code>true</code>. Due to the short-circuit evaluation, the second <code>with</code> statement is ignored and the parser never detects that <code>BAR</code> is undefined: <pre><or> <with variable="FOO"> <equals value="enabled"/> </with> <with variable="BAR"> <equals value="enabled"/> </with> </or></pre> However, the following code evaluates to <code>false</code> because the parser first encounters the access to the variable <code>BAR</code>, immediately exists and evaluates the complete expression to <code>false</code>: <pre><or> <with variable="BAR"> <equals value="enabled"/> </with> <with variable="FOO"></pre>

	<pre> <equals value="enabled"/> </with> </or> </pre>
Workaround:	Do not access undefined variables.
ID:	TRESOS-3124

Summary:	Removing mandatory system models for a disabled module does not work
Description:	If you disable a module which requires a specific system model in the project, you cannot remove the system model.
Workaround:	Temporarily remove the module from the project without deleting the configuration.
ID:	TRESOS-3126

Summary:	The ASPath pointer in XDM files erroneously allows to define a relative AUTOSAR SHORT-NAME path
Description:	If you are using ASPath path expressions in XDM files, it is currently possible to use relative paths, e.g. ASPath:DcmDslBuffer_Rx_Func. This functionality is not deliberate and can be removed in the future. The ASPath path expressions are intended for AUTOSAR compatible paths and the relative path concept of AUTOSAR works in a completely different way.
Workaround:	No workaround is available, do not use relative paths in this way.
ID:	TRESOS-3152

Summary:	The <i>Calculate Derivable Values</i> wizard is executed for disabled modules
Description:	If you disable a module, this has no effect on the <i>Calculate Derivable Values</i> wizard. This means that the wizard also overwrites the parameters of disabled modules.
Workaround:	No workaround is available.
ID:	TRESOS-3223

Summary:	The parameters of macro functions in the template-based code generator are treated identically to global variables
Description:	If you define a macro with a parameter, the parameter is treated like a global variable. This means that if you defined a variable with the same name, changes to the parameter of the macro also change the global variable.
Workaround:	Make sure not to use the same name for macro parameters and global variables, e.g. by using a prefix for parameters.
ID:	TRESOS-3249

Summary:	Ecu Config Import selecting single Config Classes does not work as expected
Description:	When using the Autosar Importer for Ecu Configurations and using one of the options <i>Replace specific config classes</i> or <i>Merge specific config classes</i> , the import does not work as expected and may lead to unexpected results. It is therefore recommended not to use these options or to carefully check the result of the import process - e.g. when you export the ECU configuration before and after the import and then compare the exported files.
Workaround:	No workaround is available.
ID:	TRESOS-3560

1.5. Incompatibilities to previous releases

Summary:	Corrections of the evaluation of attributes with type Range
Description:	In versions prior to 12.1 of EB tresos Studio, the interpretation of XDM attributes with the type <code>Range</code> was corrupt. This led to wrong or even missing range checks. This problem is fixed for EB tresos Studio 12.1.0. Please check the EB tresos Studio developer's guide for the detailed description of <code>Range</code> type attributes.
Workaround:	No workaround is available.
ID:	TRESOSWP-1958

Summary:	The option <code>Opt.OPT_USE_NODE_OPERATION</code> has been removed from the RCtxt of the Resource Public API for EB tresos Studio 12.0.0
Description:	The option <code>Opt.OPT_USE_NODE_OPERATION</code> was used to wrap changes on a model in operations, which could be undone. Since the RCtxt is not working on a project-data-model, this option was superfluous. Therefore, the option <code>Opt.OPT_USE_NODE_OPERATION</code> has been removed from the RCtxt of the Resource Public API. This leads to a better performance and a clean undo/redo stack.
Workaround:	To ensure that your code works correctly, check your Java code for the usage of the option <code>Opt.OPT_USE_NODE_OPERATION</code> and compare your expected result with the behavior described above.
ID:	TRESOSWP-1715

Summary:	There have been incompatible changes of the Public API between EB tresos Studio 2008.b and 2009.a.
Description:	The <code>ECUConfigContext</code> API class in the Java package <code>dreisoft.tresos.guided-config.api.context</code> changed between EB tresos Studio 2008.b and EB tresos Studio 2009.a.

	▶ The attribute <code>relVersion</code> now returns an instance of the Java class <code>ProjectVersion</code> instead of <code>ModuleVersion</code>. ▶ The attribute <code>generationPath</code> now returns an instance of the class <code>java.io.File</code> instead of <code>java.lang.String</code>. In the <code>plugin.xml</code> file, the equations that control the visibility of wizards changed between EB tresos Studio 2008.b and EB tresos Studio 2009.a. The changes concern how to make wizards visible in the Sidebar view and the Auto Configure toolbar. To make a wizard visible only if a project has an ECU configuration, set the property <code>ECUConfigContext</code> to <code>true</code> in the visibility equation of the extension-point of the trigger that registers the wizard.
Workaround:	Adapt your Java code and the <code>plugin.xml</code> file in the way described above.
ID:	TRESOSWP-791

Summary:	The <code>setEnabled(...)</code> methods of the Public API class <code>dreisoft.tresos.data-model2.api.model.Any</code> only work for optional nodes since EB tresos Studio 10.0.
Description:	From EB tresos Studio 10.0 on, the methods <code>setEnabled(boolean)</code> and <code>setEnabled(String, boolean)</code> of the Public API class <code>dreisoft.tresos.data-model2.api.model.Any</code> only have an effect on optional nodes. This is so, because the enablement state of non-optional nodes cannot be changed. Therefore non-optional nodes are now ignored by these methods. If you call one of these methods for a non-optional node, the consequence is as follows: ▶ If the current enablement of the node equals the enablement provided as parameter to the method, the Boolean return value will be <code>true</code>. ▶ If the current enablement of the node does not equal the enablement provided as parameter to the method, the Boolean return value will be <code>false</code>.
Workaround:	Check your Java code especially when invoking the named methods on non-optional nodes. The return value may differ compared to executions of the code with older EB tresos Studio versions. The same applies to calls of the <code>isEnabled()</code> or <code>isEnabled(String)</code> methods after calling one of the <code>setEnabled(...)</code> methods.
ID:	TRESOS-2268

Summary:	The behavior of the methods <code>equals(Object)</code> of the Public API class <code>dreisoft.tresos.launcher2.api.module.EcuType</code> has been changed
Description:	Prior to EB tresos Studio 10.0, the implementation of the method <code>equals(Object)</code> in the class <code>dreisoft.tresos.launcher2.api.module.EcuType</code> did not comply to the common Java coding guidelines. The current implementation now ensures that an <code>EcuType</code> equals the given <code>java.lang.Object</code> only, if: ▶ the <code>EcuType</code> and the given <code>Object</code> both are the same object, or

	▶ the given <code>Object</code> is not <code>null</code>, and ▶ the given <code>Object</code> is an instance of <code>dreissoft.tresos.launcher2.api.module.EcuType</code>, and ▶ the target (see <code>EcuType.getTarget()</code>) strings of both <code>EcuTypes</code> are equal, and ▶ the derivative (see <code>EcuType.getDerivate()</code>) strings of both <code>EcuTypes</code> are equal.
Workaround:	To ensure that your code works correctly, check your Java code for the usage of the named method and compare your expected result with the behavior described above.
ID:	TRESOS-2455

Summary:	Limited support for AUTOSAR 3.2.1 since EB tresos Studio 13.1.0
Description:	Due to the implementation of AUTOSAR 3.2.2, the support for AUTOSAR 3.2.1 had to be reduced. Starting with EB tresos Studio 13.1.0, the import of AUTOSAR 3.2.1 ECU configuration files is no longer possible with validation of the input data turned on.
Workaround:	Since there are large similarities between AUTOSAR 3.2.1 and AUTOSAR 3.2.2 ECU configuration files, the import can usually be done with validation turned off. In this case, the data is not validated against the AUTOSAR XML schema. The importer tries its best to import that data into the 3.2.2 model, but might not be able to import all data.
ID:	TRESOSWP-2159

Summary:	Warning ID for unresolved references in AUTOSAR system description files has changed
Description:	The ID of the warning that is reported if an AUTOSAR system description file contains references that can not be resolved has changed for some use cases. While in earlier releases of EB tresos Studio <code>SYSDI_204</code> was reported in all cases, this partly has changed to <code>TRESOSDB_75</code> .
Workaround:	If you do not want to receive warnings for open references in AUTOSAR system description files, please suppress the message ID <code>TRESOSDB_75</code> in addition to <code>SYSD_204</code> now.
ID:	TRESOS-3510

1.6. Changes in version 14.2.1

1.6.1. Fixed problems

Summary:	Dependencies between module configuration plugins are now considered correctly
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Description:	Since EB tresos Studio 14.1.0 you can define dependencies between your module plugins as a module developer. If a read-only ECU configuration parameter with a calculated default value refers to another module definition (e.g. when using subderivatives with ECU resource properties), EB tresos Studio may have calculated those default values incorrectly in certain cases when the project was loaded. This problem is fixed now.
ID:	TRESOS-3520

Summary:	Fixed processing of <code>SwDataDefProps</code> elements in System Model Upgrade
Description:	In previous versions of EB tresos Studio the system model upgrade incorrectly moved source model AUTOSAR 3.x <code>SwDataDefProps</code> elements into arbitrary <code>VariableDataPrototypes</code> in the generated AUTOSAR 4.0 model. This happened if the <code>SwDataDefProps</code> were contained in <code>RecordElements</code> , <code>ArrayElements</code> , or <code>InterRunnableVariables</code> . If the <code>VariableDataPrototypes</code> already contained an <code>SwDataDefProps</code> element, the upgrade was terminated and an error was issued. This problem is fixed now.
ID:	TRESOS-3620

Summary:	Fixed processing of <code>UUID</code> values in System Model Upgrade
Description:	In previous versions of EB tresos Studio the System Model Upgrade incorrectly duplicated <code>UUID</code> attributes of several AUTOSAR 3.x source model elements. This resulted in an error if the <i>improved merge algorithm based on UUID matching</i> was selected in an import configuration for a Fibex, LDF, or AUTOSAR 3.x system description file. This problem is fixed now.
ID:	TRESOS-3623

Summary:	Fixed handling of initial values and invalid values of signals during System Model Upgrade
Description:	In earlier versions of EB tresos Studio, the System Model Upgrade from AUTOSAR 3.x to 4.0 did not support opaque literals as invalid and initial values of signals. This restriction has been removed.
ID:	TRESOS-3582

1.6.2. Changes/new features

1.7. Changes in version 14.2.0

1.7.1. Fixed problems

Summary:	Undo was not possible after changing a value within a table in the Generic Configuration Editor
Description:	In EB tresos Studio version 14.1, after you changed a value within a table cell in the Generic Configuration Editor, you could not revert the change by pressing the Undo button. This bug is now fixed.
ID:	TRESOS-3571

Summary:	An error message is shown when you try to rename a project to an invalid name
Description:	In previous versions of EB tresos Studio, you could rename a project to an invalid name. An invalid name is a name that is not a valid AUTOSAR SHORT-NAME, e.g. a name that contains spaces. In that case the project was not fully usable and e.g. displayed errors during code generation. This is fixed now. If you try to rename a project to an invalid name, an error message is shown and you cannot complete the rename action.
ID:	TRESOS-1119

Summary:	No more outdated entries in tables of the Generic Configuration Editor after the execution of a guided configuration wizard
Description:	In previous versions of EB tresos Studio, sometimes multiple entries with the same index value were visible in the Generic Configuration Editor after you executed a guided configuration wizard. This is fixed now.
ID:	TRESOS-3488

Summary:	The display of Published Information parameters and containers has been fixed
Description:	In previous versions of EB tresos Studio, containers with a name that ends in <code>PublishedInformation</code> were rendered incorrectly. Previously only the first element of lists was shown and all parameters were shown on the <i>Published Information</i> tab. This has been fixed now. The items with a name that ends in <code>PublishedInformation</code> are now treated equally to normal editor items with respect to the location in the editor. Also all elements of lists are displayed. Only the container named <code>CommonPublishedInformation</code> is treated specially. This is required by AUTOSAR.

ID:	TRESOS-3154
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Summary:	Module Configuration Editor only shows valid reference targets
Description:	In previous releases of EB tresos Studio, the Module Configuration Editor offered invalid reference targets for references that point into a choice container. This led to subsequent errors during model validation, if such an invalid node was selected. This problem is fixed now. Only valid reference targets are shown now.
ID:	TRESOS-3307

Summary:	<i>Compare</i> editor for XDM files no longer hangs up
Description:	In previous releases of EB tresos Studio, the EB tresos Studio GUI may have hung up when you compared XDM files with the <i>Compare</i> editor for XDM files which contained lists with many nodes that were moved to another position. This is fixed now.
ID:	TRESOS-3406

Summary:	The XPath function <code>substring()</code> logs a warning message if it is used with 0 as start position
Description:	If you use the XPath function <code>substring()</code> with three arguments and provide 0 as start position, EB tresos Studio now displays a warning message every time the function is executed. The function is not defined for start position 0 in the XPath standard. In the implementation provided by EB tresos Studio the function returns a result but the result is clearly wrong because it is shorter than the specified length. In previous releases of EB tresos Studio, the identical value was returned, but no warning was shown.
ID:	TRESOS-3242

Summary:	<code>tresos_cmd.bat legacy</code> command line does not report erroneously unresolved references any more
Description:	In EB tresos Studio 14.1.0, the <code>tresos_cmd.bat legacy</code> command line under certain circumstances reported that references defined in AUTOSAR system description files could not be resolved although the references were resolved. Additionally, warnings about unresolved references may have been reported multiple times. This also was the case for the falsely reported ones stated above. Those problems are fixed now. EB tresos Studio now only reports actually unresolved references once.

ID:	TRESOS-3509
Summary:	Resolved delays before code generation
Description:	In EB tresos Studio 14.1.0, configuration code generation started with a significant delay when WibuKey dongles were employed for license authentication and a large number of module configurations were generated in parallel. The causes for these delays have been removed now.
ID:	TRESOS-3548
Summary:	Users' choice is now persisted for Do not show this dialog again confirmation dialogs
Description:	In previous releases of EB tresos Studio for several actions confirmation dialogs are shown with <i>Do not show this dialog again</i> check box. This confirmation dialog appears e.g. when you add or remove parameter comments or when you try to close locking parts automatically. Users could choose to not show this dialog again and whether to perform or deny the action. But the users' decision was not persisted and when you executed the same action again, the default answer was performed. Now the check box is named <i>Remember my decision</i>, the users' choice is persisted and used for all further executions of the same action. The decision can be revoked in the EB tresos Studio preferences.
ID:	TRESOS-2668

1.7.2. Changes/new features

Summary:	Configurable multiple configuration support for AUTOSAR 4.1.1 post build scenarios
Description:	EB tresos Studio now features enhanced support for the Multiple Configuration Container (MCC) approach for post build ECU configuration scenarios as specified by AUTOSAR 4.1.1. With the help of a specific post build configuration management module, you can now aggregate selected MCCs of your project into named MCC sets and select one of those sets to be the current working set. After you switch such a configuration project into post build only mode, EB tresos Studio on the one hand will perform additional checks of the consistency of the configuration data in the different MCCs. On the other hand, EB tresos Studio will hinder you from changing certain ECU configuration parameters that were already configured. For details, please refer to the <i>EB tresos Studio new and noteworthy</i> document.

ID:	TRESOSPF-44
Summary:	New rules implemented for the legacy vsmdcheck command regarding AUTOSAR 4.1.1 post-build
Description:	EB tresos Studio 14.2 provides new rules for checking Vendor Specific Module Definitions (VSMD) against the refined AUTOSAR 4.1.1 specification for post-build capable modules. The new rules cover the following AUTOSAR constraints and requirements: <code>[constr_5500]</code>, <code>[TPS_ECUC_02132]</code>, <code>[TPS_ECUC_06051]</code>, <code>[TPS_ECUC_08005]</code> and <code>[TPS_ECUC_08007]</code>. Additionally EB tresos Studio 14.2 provides a new rule set named <code>asc:4.0.3.RFCs</code> which complements the existing <code>asc:4.0.3</code> ruleset according to the changed AUTOSAR requirements.
ID:	TRESOSPF-91
Summary:	Post-build verification of function names ([constr_5502])
Description:	In compliance to the AUTOSAR constraint <code>[constr_5502]</code> , EB tresos Studio now also verifies that you do not accidentally introduce new function names during post build time, and creates entries in the <i>Problems View</i> accordingly.
ID:	TRESOSWP-2532
Summary:	Selectable AUTOSAR XML schema variant in <i>System Description Importer</i>
Description:	For AUTOSAR releases that come with different XML schema variants (e.g. AUTOSAR 4), you may now choose the AUTOSAR schema variant that EB tresos Studio shall use to validate the XML file when you import system data with the <i>System Description Importer</i>. In previous releases, this choice was not available and the strict schema supplied by AUTOSAR was always used.
ID:	TRESOSPF-30
Summary:	Limitations section cleaned up
Description:	The <i>Limitations</i> section of the EB tresos Studio release notes has been cleaned up. Some long open limitations have been moved into the matching places in the EB tresos Studio user's guide or developer's guide.
ID:	n/a
Summary:	Workflows view is now project-specific
Description:	In previous releases of EB tresos Studio, the <i>Workflows</i> view had one globally saved state. This was problematic when users performed one single workflow for several projects. They did not know anymore where they interrupted their work for each project.

	Now the <i>Workflows</i> view is project-specific and the state information is stored in each project. When users select the open project now, the state information gets restored and is displayed in the <i>Workflows</i> view.
ID:	TRESOSPF-42

Summary:	Parameters within optional containers are displayed now in GUI tables
Description:	In previous releases of EB tresos Studio, the parameters inside an optional container could only be seen in the <i>Generic Editor</i>. Now the tables in the GUI also display these parameters as columns. If the container element is disabled, the parameters are not editable. The parameters inside optional containers also appear in the <i>Column Selection Dialog</i> and the <i>Bulk Change Editor</i>.
ID:	TRESOSWP-2599

Summary:	Enhancements for Bulk Change Editor - displaying optional parameters
Description:	In previous releases of EB tresos Studio, the Bulk Change was possible only for configuration parameters that were not optional, were not in a subcontainer (list in list), and were visible as column in the corresponding table. Bulk Change now also supports enabling and disabling optional parameters. The Bulk Change of all parameters in the container including optional parameters is possible. The parameters inside optional containers are also visible in the Bulk Change Editor, however their modification is only possible if the parent container is enabled.
ID:	TRESOSPF-32

1.8. Changes in version 14.1.0

1.8.1. Fixed problems

Summary:	Dependencies between module configuration plugins are now considered
Description:	In former releases of EB tresos Studio, if you created an ECU configuration project with modules, where one module definition contained a read-only parameter with a calculated default value that referred to another module definition (e.g. when using subderivatives with ECU resource properties), EB tresos Studio may not have calculated the value correctly. EB tresos Studio now considers any defined dependencies between the Eclipse bundles that contain the module definitions when calculating such parameter values.

	If you are a module developer, and your module refers to some other module for calculating default values of read-only parameters, please make sure to define this dependency in the MANIFEST.MF file of your module plugin (within the <code>Require-Bundle: manifest</code> header).
ID:	TRESOS-3318

Summary:	API methods <code>DataModelAccess.getEntityBy[AS]Path</code> failed
Description:	In former releases of EB tresos Studio, if there was ECU configuration data within an AUTOSAR package with the same SHORT-NAME as a package within the system model (tresosDB), the API methods <code>DataModelAccess.getEntityByPath</code> and <code>DataModelAccess.getEntityByASPath</code> failed to retrieve any system model entities from within that package. This is fixed now.
ID:	TRESOS-3427

Summary:	It is now possible to edit SHORT-NAMEs of choices in lists
Description:	In former releases of EB tresos Studio, pages in the <i>Generic Configuration Editor</i> via which you can edit the choice container of a choice in a list did not let you change the SHORT-NAME of the choice. This is fixed now.
ID:	TRESOS-1931

Summary:	The name of top level container was editable in the Generic Configuration Editor
Description:	In former releases of EB tresos Studio, e.g. after you selected another tab in the Generic Configuration Editor, the name textfield for the toplevel container became editable although it should stay read-only. This is fixed now.
ID:	TRESOS-3462

Summary:	API class <code>CodeGeneratorContext</code> lost variable values between generation steps
Description:	In EB tresos Studio 14.0, the API class <code>dreisoft.tresos.generator.api.generator.CodeGeneratorContext</code> did not preserve custom variables between generation steps (e.g. added by <code>CodeGeneratorContext.setVariable(String, Object)</code>). This is fixed now.
ID:	TRESOS-3418

Summary:	Error if comparing XDM files in the <i>Commit</i> dialog
Description:	In former releases of EB tresos Studio, if you double clicked on an XDM file to see its changes within the <i>Commit</i> dialog, an error message appeared and no differences were shown. This is fixed now.
ID:	TRESOS-3447

Summary:	Improved performance of Generic Configuration Editor when adding many elements to a list
Description:	In former releases of EB tresos Studio, when you added many new elements (e.g. 2.000) to a list at once while the Generic Config Editor was currently showing the table, e.g. by using a custom Guided Configuration dialog, EB tresos Studio hung up for some time (e.g. 5 minutes). This is fixed now. EB tresos Studio will now be responsive again after a few seconds in the given example.
ID:	TRESOS-3435

Summary:	The EB tresos Studio Public API Java documentation is now visible in the Eclipse Help
Description:	In former releases of EB tresos Studio, the EB tresos Studio Public API Java documentation was not visible in the the Eclipse Help, after you had set up the Eclipse IDE as described in the EB tresos Studio developer's guide. This is fixed now.
ID:	TRESOS-3424

Summary:	Guided configuration wizard IDs with underscore characters are now handled without error
Description:	In former releases of EB tresos Studio, the tool displayed errors to the Error Log if you used underscore characters in the ID of the guided configuration wizard in the file plugin.xml. This is fixed now.
ID:	TRESOS-3433

Summary:	Solved duplicate short names when upgrading AUTOSAR 3.2 system description files
Description:	Previous releases of EB tresos Studio created duplicate short names for the elements <code>ResourceConsumption</code> and <code>Code</code> when you upgraded AUTOSAR 3.2 system description files which contained <code>Implementation</code> elements having the same short name as the aggregated <code>Code</code> element. This issue has been solved as the short name of each <code>ResourceConsumption</code> element is suffixed with <code>"_resourceConsumption"</code>.
ID:	TRESOS-3448

Summary:	Conversion from XMD to EPD for AUTOSAR 3.2 files fixed
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Description:	In previous releases of EB tresos Studio, the writing of epd files stopped with an <code>Null-Pointer</code> error message for AUTOSAR version 3.2 if the input file contained <code>ADMIN-DATA</code> elements. This is now fixed and AUTOSAR 3.2 files with the <code>ADMIN-DATA</code> tag can be written.
ID:	TRESOS-3440

Summary:	Choice container tabs are displayed in the available editor window
Description:	In previous releases of EB tresos Studio, the tabs displayed in the editor for choice containers were not stacked and thus made the editor window as wide as needed to display all tabs. If there were many choice options, scrollbars appeared for the editor view. In some cases, as e.g. in the BswM module, this could lead to subsequent problems, e.g. unusable drop-down boxes for references because of an extremely large editor width. This is now fixed and the tabs of choice containers only take up the available width. If the width is not enough for all tabs, the tabs are stacked and a button appears that displays the remaining tabs as a list.
ID:	TRESOS-3444

Summary:	Copy to clipboard is now possible in the <i>Error Log</i> and <i>Problems</i> view dialogs
Description:	The detail dialogs for <i>Error Log</i> and <i>Problems</i> view entries now display a button with the label <i>Copy to Clipboard</i> which enables you to copy the entry into the system clipboard.
ID:	TRESOS-3451

Summary:	Fibex support in 64 bit mode has been added
Description:	EB tresos Studio now provides the possibility to import Fibex files also in its 64 bit mode.
Workaround:	Import Fibex files in 32 bit mode.
ID:	TRESOSWP-2364

Summary:	Fibex support for conditional timings has been added
Description:	It is now possible to import multiple CYCLIC-TIMINGS and EVENT-CONTROLLED-TIMINGS which contain START/STOP/SEND-CONDITIONs and SIGNAL-TRIGGERs.
ID:	TRESOS-3348

Summary:	The <i>Error Log</i> view now displays all entries
Description:	In previous releases of EB tresos Studio not all messages were shown in the <i>Error Log</i> view. If you had enabled the option <i>Clear error log view on startup</i> , the <i>Error Log</i> view had

	not been cleared during startup of Studio. Instead, it was cleared after the initialization of the error log, which happens when you click the <i>Error Log</i> view tab and make it visible. This means that if a EB tresos Studio component reported an error message, then ▶ the error log was not marked as modified because the title was not rendered in bold font ▶ messages that had been logged before you opened the log view were not visible This is now fixed and all messages are displayed in the <i>Error Log</i> view even if you did not open it. Along with the fix, some speed optimizations are introduced and the maximum amount of log entries displayed is increased to 100.000 entries.
ID:	TRESOS-3360

1.8.2. Changes/new features

Summary:	Improved merge behavior of System Description Importer
Description:	Besides the already known (legacy) merge algorithm of the system description importer, a new improved merge algorithm is now provided. You can still decide which merge algorithm to use. For newly created system description importers, the new merge algorithm is used by default. The new merge algorithm is based on the atpSplitable concept of AUTOSAR. The AUTOSAR meta model stereotypes <splitable> (ASR 3.x) / <atpSplitable> (ASR 4.x) define which elements can be split up over several physical files. The new improved merge algorithm merges the input files accordingly. Additionally, the new merge algorithm uses UUID matching to find e.g. renamed elements. For example, you imported a file into the existing project model and rename an element in the file afterwards. If the element still has the same UUID, the merge algorithm will find the correct entity and rename it in the project model when you import the file again using the new algorithm.
ID:	TRESOSWP-2151

Summary:	Import order of input files inside the System Description Importer can now be specified
Description:	In previous EB tresos Studio versions, the list of files was sorted alphabetically and could not be changed. The system description importer now provides the possibility to change the import order of the selected input files. Newly added files are always inserted at the end of the list. With the new Up and Down buttons, it is now possible to change the order of the input files. It is only possible to move exactly one file at a time.

	Through the import order, you can specify the priority of the files. When you import several files containing the same element, the import order is important to specify which file has the highest priority while it is being imported. The topmost file in the list always has the highest priority.
ID:	TRESOSWP-2118
Summary:	Support to edit the <code>SHORT-NAME</code> of containers with upper multiplicity=1
Description:	You can now edit the <code>SHORT-NAME</code> of containers which have up to one occurrence (including optional containers) using the Generic Configuration Editor.
ID:	TRESOSWP-2441
Summary:	Undo/Redo buttons are now configurable for guided configuration dialogs
Description:	For guided configuration dialogs by default Undo/Redo buttons are shown on top of the dialog. With the newly added optional attribute <code>undoRedo</code> of extension point <code>dreisoft.tresos.guidedconfig.api.plugin.trigger</code> it is now possible to display the dialog without Undo/Redo buttons. If the attribute is not set, then by default the Undo/Redo buttons are displayed. This attribute is only relevant for triggers with <code>wizardType="dialog"</code> .
ID:	TRESOSWP-1874
Summary:	Support for <code><atpSplitable></code> stereotype during TresosDB merge
Description:	EB tresos Studio now can use the <code><atpSplitable></code> stereotypes from the AUTOSAR meta model when merging TresosDBs. This allows to merge arbitrary system models in a generic way.
ID:	TRESOSWP-2410
Summary:	Parallel code generation
Description:	In former versions of EB tresos Studio all code generators were executed sequentially. Now code generators are launched in parallel if this is supported by the generators.
ID:	TRESOSWP-2407
Summary:	Parallelization of configuration verification
Description:	The verification of the ECU configuration values was done sequentially in former versions of EB tresos Studio. This task is parallelized now to improve performance.
ID:	TRESOSWP-2320
Summary:	The window title now shows more information
Description:	The window title now shows the product version, the workspace directory and the installation directory.
ID:	TRESOSWP-2447

1.9. Changes in version 14.0.0

1.9.1. Fixed problems

Summary:	Not able to reference sub-package nodes
Description:	In previous release, references always created a warning, if the reference-target is contained within a module containing sub-packages in case the referencable nodes are defined by using the path-prefix "/AUTOSAR/EcuDefs". This is fixed now.
ID:	TRESOS-3368

Summary:	Performance of system model description import
Description:	When importing a system model description, the performance was very slow if an ecu extract has been created. This part of the import process is 6 times faster now.
ID:	TRESOS-3292

Summary:	False warnings about inconsistent patch versions of modules
Description:	In previous versions, under certain circumstances EB tresos Studio reported false warnings like <code>CommonPublishedInformation</code> parameter <code>"ArPatchVersion"</code> default in schema XDM of module <code>"<SomeModule>"</code> does not match module registration in its <code>plugin.xml</code> . Fixed value in schema. " This is now fixed and such warnings are not reported falsely any more.
ID:	TRESOS-3022

Summary:	Import of DBC files with messages containing signals and/or signal groups with equal names
Description:	In previous versions of EB tresos Studio, the DBC Importer issued an error message if signals and/or signal groups with equal names were encountered in one and the same CAN message. Such name conflicts are now resolved by adding suffixes to the signal and signal group names.
ID:	TRESOS-3376

Summary:	Import of Fibex files containing references to custom schema files
Description:	In previous versions of EB tresos Studio, the Fibex importer reported a parser error when Fibex files, which contained references to custom sub schema files, were imported. When you turn off the option <i>Validate against XML schema</i> , you can import such files now.
ID:	TRESOSWP-2337, TRESOS-3199

Summary:	Incorrect display of errors and warnings in <i>Problems</i> view when loading several projects at once is fixed now
Description:	In previous versions of EB tresos Studio, the <i>Problems</i> view did not show errors and warnings correctly when you loaded several projects at once. All errors and warnings were shown, but accumulated in the same project node. This issue is fixed now.
ID:	TRESOS-3310, TRESOS-3354

Summary:	EB tresos Studio now reports warnings if large integral values cannot be imported from ARXML files without a loss of precision
Description:	In previous releases of EB tresos Studio, when you imported AUTOSAR XML files which contained overly large numbers (e.g. of AUTOSAR primitive type <i>Numerical</i>), EB tresos Studio did not inform you about a loss of precision. This loss of precision occurred when you transformed the string value into the internal IEEE-754-1985 double-precision format. Now you will receive a warning message like the following in the case of the loss of precision: The value "9223372036854775806" of the attribute MAX can not exactly be represented in the internal format and will be rounded to "9.223372036854776E18".
ID:	TRESOS-3369, TRESOS-3306

Summary:	Auto-detection of AUTOSAR version failed for UTF-16 or UTF-32 encoded files
Description:	Previous versions of EB tresos Studio sometimes failed to auto-detect the AUTOSAR version of unicode ECU configuration files and system description files. This problem occurred for AUTOSAR files which used the encoding UTF-16 or UTF-32 containing a <i>byte order mark (BOM)</i> . This problem has been resolved.
ID:	TRESOS-3356

Summary:	Create ECU extract dialog failed to create the expected ECU extract
Description:	In previous releases of EB tresos Studio, you could not use the <i>Create ECU extract</i> dialog to create the ECU extract for the system model, when there was more than one system imported which had ECU instances with equal SHORT-NAMES. This is fixed now.
ID:	TRESOS-3331

Summary:	No configuration XDM file after re-adding a module configuration using the project <i>Properties</i> dialog
Description:	In previous releases of EB tresos Studio there was a problem when you used the <i>Modules</i> page of the project <i>Properties</i> dialog. If you removed an existing module configuration together with its XDM file and re-added it again without applying the changes in be-

	tween, the XDM file did not exist afterwards. This problem only appeared when you had opened the page <i>Configuration Project</i> of the project <i>Properties</i> dialog first. This is fixed now.
ID:	TRESOS-3320

Summary:	Subpackages support did not work in legacy generate mode
Description:	In previous releases of EB tresos Studio, an error was reported when you tried to run code generation for a module with subpackages. The error indicated that the MODULE-DEFINITION could not be found. This is fixed now.
ID:	TRESOS-3311

Summary:	Foreign references are now correctly converted
Description:	In previous releases of EB tresos Studio, if you derived a VSMD from a StMD with use of the <code>legacy vsmd</code> command from the command line, foreign references were not converted correctly. Thus these foreign references may have lead to wrong warning messages in the problems view such as like <i>The node "... may not refer to the node "...</i>. Additionally, the user could not choose the reference from a list of existing targets because the references drop box stayed empty. This is fixed now. If you are a module developer and your module uses foreign references, please convert the StMD again and merge the differences into your VSMD module schema XDM file.
ID:	TRESOS-3215

Summary:	EB tresos Studio ignored the <code>outputDir</code> option for <code>ng.extgen</code> generators
Description:	In previous releases EB tresos Studio ignored the <code>outputDir</code> option for code generators that used the <code>ng.extgen</code> Ant task of the NG Generator API. This regards the Java API method <code>dreisoft.tresos.launcher2.api.project.Project.generate(String, Map<GenerateOption, String>, APIOperationStatus)</code> as well as the command line option <code>-o <Output-Path></code> . This bug is fixed now.
ID:	TRESOS-3295

Summary:	Fixed hang up during start
Description:	In previous releases EB tresos Studio sometimes did not start and hung up while showing the splash screen without showing any further progress. This is fixed now.
ID:	TRESOS-3143

Summary:	Fixed performance issue with progress dialog
Description:	When using EB tresos Studio under Windows 7, performance issues occurred, when EB tresos Studio showed up progress dialogs. Those are fixed now.
ID:	TRESOS-3187

Summary:	Progress dialogs open with some delay to avoid flickering in the GUI
Description:	For long running actions, the progress dialog is now displayed after some delay. This reduces flickering in the GUI.
ID:	TRESOS-2626

Summary:	Instance references are now correctly converted
Description:	In previous releases of EB tresos Studio, if you derived a VSMD from a StMD using the <code>legacy vsmd</code> command from the command line, instance references were not converted correctly. Thus these instance references may have lead to wrong warning messages in the problems view such as <i>The node "..." may not refer to the node "..."</i>. Additionally, the user could not choose the reference from a list of existing targets because the references drop box stayed empty. This is fixed now. If you are a module developer and your module uses instance references, please convert the StMD again and merge the differences into your VSMD module schema XDM file.
ID:	TRESOS-3232

Summary:	Service component generation in Rte Editor no longer fails if a project with AUTOSAR version <code>mixed</code> is used
Description:	In previous releases of EB tresos Studio, if you used a project with AUTOSAR version <code>mixed</code> and AUTOSAR 3.x modules, the service component generation in the Rte Editor failed with an error message. This is fixed now.
ID:	TRESOS-3175

Summary:	ECU-local system descriptions are handled correctly
Description:	In previous releases of EB tresos Studio, ECU-local system descriptions (<i>system extracts</i>) were not handled correctly in AUTOSAR 4.0 projects. All delegation connector chains leading to the outside of the top level composition were ignored. This problem is fixed now.
ID:	TRESOS-3137

Summary:	Invalid <code>SHORT-NAME</code> input now leads to an error marker
Description:	In previous releases of EB tresos Studio, when you set the <code>SHORT-NAME</code> of a node in the Generic Configuration Editor to some invalid value, the change was ignored and EB tresos Studio logged an error in the <i>Error Log</i> view. An invalid value is for example the use of a trailing blank character. Now EB tresos Studio shows the invalid name with an error marker instead and reports the invalid <code>SHORT-NAME</code> in the <i>Problems View</i>.
ID:	TRESOS-3150

Summary:	Fixed Sidebar view when displaying several triggers with the same path inside the same category
Description:	In previous releases of EB tresos Studio, there has been a problem if several triggers have been registered with the same path inside the same category of the <i>Sidebar</i> view. In this case, if one of these triggers was not visible due to the visibility restriction in the <code>plugin.xml</code> , all other triggers with the same path were invisible, too. This behavior is fixed now.
ID:	TRESOS-3129

Summary:	No rule violation 6051 is reported incorrectly any more
Description:	In previous releases of EB tresos Studio, if you implemented a VSMD which only supports one configuration variant, the <code>vsmdcheck</code> command incorrectly reported the VSMD <i>rule violation 6051</i> for the node <code>IMPLEMENTATION_CONFIG_VARIANT</code> . This problem is fixed now.
ID:	TRESOS-3111

Summary:	Fixed exception in system model upgrade when opaque literals are processed
Description:	Previous releases of EB tresos Studio failed to upgrade AUTOSAR 3.2 system description files containing opaque literals to AUTOSAR 4.0 and issued an error message as follows: "cannot DETACH database within transaction". The processing of opaque literals has been fixed.
ID:	TRESOS-3322

Summary:	Fibex Importer now does not write ARLimits in case of an overflow
Description:	In previous releases of EB tresos Studio, the value of ARLimits created for LOWER/UPPER-LIMITs of PHYS-CONSTR was wrong if the conversion to the coded value resulted in a value exceeding the largest valid int64 value. In case of such an overflow no ARLimit is now created at all.
ID:	TRESOS-3345

Summary:	The <i>Results</i> view now shows removed nodes
Description:	When a guided config wizard removes <code>DCtxt</code> nodes, they are now shown in the <i>Results</i> view.
Workaround:	
ID:	TRESOS-3267, TRESOS-2456

Summary:	Existing attributes of parameters are lost during ECU configuration import
Description:	In previous releases of EB tresos Studio, under some circumstances, existing attributes of parameters were lost when importing the ECU configuration via DBC Importer, LDF Im-

	porter, FIBEX Importer, System Description Importer or tresosDB Importer. Now the problem is fixed and the attributes of the existing and imported configuration are merged correctly.
ID:	TRESOS-3343, TRESOS-3363

1.9.2. Changes/new features

Summary:	Performance optimizations
Description:	Several performance optimizations have been done, for example when accessing the internal datamodel via XPath. This affects a lot of operations, e.g. code generation, epc file export and verifying the configuration values. An overall performance improvement of 13% has been measured using internal tests.
ID:	TRESOSWP-2265

Summary:	64 bit support
Description:	It is now possible to run EB tresos Studio as a 64 bit application by simply using tresos_gui_64.exe for gui mode or tresos_cmd_64.bat for command line mode. This is especially useful for huge projects where the maximum usable memory of a 32 bit application is restricted and therefore might not run properly.
ID:	TRESOSWP-2356

Summary:	Specific menu items for generator modes
Description:	EB tresos Studio is now able to show the menu items for specific generator modes which use translatable labels and specific icons, instead of just showing the symbolic name of the generator mode. For further details refer to the description of the new <code>modeMenuItem</code> element of the <code>dreisoft.tresos.generator.api.plugin.generator</code> extension point.
ID:	TRESOSWP-2319

Summary:	Eclipse platform updated to version 3.8
Description:	The platform on which EB tresos Studio is developed has changed with this release. EB tresos Studio is now based on the Eclipse 3.8.0 platform.
ID:	TRESOSWP-2225

Summary:	Error Log view now supports Ctrl + C
Description:	In EB tresos Studio it is possible to copy the selected <i>Error Log</i> entry to the clipboard via the context menu Copy . Now it is also possible to use keyboard shortcut Ctrl + C .
ID:	TRESOSWP-543

Summary:	New dialog when re-adding a module configuration
Description:	When you delete an existing module configuration together with its configuration file, and re-add the same module configuration without applying the change in between, EB tresos Studio will now prompt you with a dialog. The dialog will ask you whether you want to re-use the still existing file or not and sets the Load existing file option accordingly. The dialog contains an option Remember my decision to make your answer the default behavior so the dialog will not appear again. You can adjust the behavior on the <i>Configuration</i> page in the <i>Preferences</i> dialog of EB tresos Studio.
ID:	TRESOSWP-2303

Summary:	Optimized runtime and memory footprint of system model
Description:	In previous releases of EB tresos Studio, all system model classes were loaded immediately when you opened the first configuration project or when EB tresos Studio automatically re-opened projects during startup. Now EB tresos Studio lazily loads those classes as required. This significantly reduces the time for opening the first project, as well as the memory footprint.
ID:	TRESOSWP-2287

Summary:	Loading of module configurations is now done in parallel
Description:	If you load a project, the loading of the module configurations is now done in parallel in multiple threads. This enables the system to use all available cores of the computer and reduces the time to load a project considerably.
ID:	TRESOSWP-2264

Summary:	Removed confirmation dialog when canceling a wizard dialog
Description:	In previous releases of EB tresos Studio, a confirmation dialog appeared when you clicked the Cancel button inside a guided configuration dialog. The confirmation dialog asked you if you really wanted to cancel or return to the dialog again. Since the wizard dialog will always be saved automatically, this confirmation dialog is not necessary and was now removed.
ID:	TRESOSWP-2179

Summary:	Reference combo boxes now use the available width in the Generic Configuration Editor
Description:	In previous releases of EB tresos Studio, reference combo boxes had a fixed width which often made it hard to identify the values, as the node paths usually are too long to fit into. Now reference combo boxes use the available width and resize when you widen the editor within the workbench.
ID:	TRESOS-3141

Summary:	Project builder for generating National Language Support (NLS) and Operation Status (OS) Java source and property files
Description:	An automatic project builder replaces the former <i>Generate NLS</i> menu item and toolbar button (part of the EB tresos Studio Eclipse tooling). For further details, refer to the EB tresos Studio developer's guide, chapter <i>The NLS/OS builder</i> .
ID:	TRESOSWP-2163

Summary:	Hyperlink support in the details dialog of the <i>Error Log</i> view.
Description:	It is now possible to follow links in the details dialog of the <i>Error Log</i> view to get directly to the node in the Configuration Editor.
ID:	TRESOSWP-2023

Summary:	Hyperlink support in the details dialog of the <i>Problems</i> view.
Description:	It is now possible to follow links in the details dialog of the <i>Problems</i> view to get directly to the node in the Configuration Editor.
ID:	TRESOSWP-2023

Summary:	Improved processing of <code>UnqueuedReceiverComSpec.invalidationPolicy</code> in System Model Upgrade
Description:	In previous versions of EB tresos Studio the system model upgrade generated AUTOSAR 4.0 models that could not be processed by the RTE if the AUTOSAR 3.x source model contained <code>UnqueuedReceiverComSpec</code> elements that yielded different <code>invalidationPolicy</code> values, but referred to the same <code>DataElementPrototype</code> . An extension was added so that in such cases the <code>invalidationPolicy</code> can now be retrieved from the <code>UnqueuedReceiverComSpec</code> elements in the upgraded AUTOSAR 4.0 model. In all other cases, the <code>invalidationPolicy</code> is retrieved from the common <code>Interface</code> element.
ID:	TRESOSWP-2334

1.10. Changes in version 13.1.0

1.10.1. Fixed problems

Summary:	The System Description Exporter no longer merges multiple instances of clusters and controllers
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Description:	For AUTOSAR 4.x, the <i>System Description Exporter</i> merged the following elements to one element each, and thus did not export their SHORT-NAME: CanCluster, CanCommunicationController, FlexRayCluster, FlexRayCommunicationController, LinCluster, LinMaster, LinSlave, TtcanCluster, TtcanCommunicationController, EthernetCluster and EthernetCommunicationController. This is fixed now.
ID:	TRESOS-3211

Summary:	Commandline call to legacy make contract reported an unknown error
Description:	If you called <code>legacy make contract</code> on the commandline, under certain circumstances the following error occurred in previous releases of EB tresos Studio: "An unknown error occurred: Unknown reference type _package" This is fixed now.
ID:	TRESOS-3170

Summary:	The rule violation 6051 is now reported correctly
Description:	If you implement a VsMD which only supports one configuration variant, the <code>vsmdcheck</code> command now correctly reports the VSMD rule violation 6051 for the node <code>IMPLEMENTATION_CONFIG_VARIANT</code> .
ID:	TRESOS-3111

Summary:	Saving a second time in the Compare Editor for loaded configuration XDM files
Description:	When you tried to save an XDM file that you edited within the Compare Editor for XDM files for a second time, and the XDM file contained a module configuration of a loaded project, a "no data available" error message was shown, and your changes were not saved. This bug is fixed now.
ID:	TRESOS-3230

Summary:	Service component generation in Rte editor no longer fails if a project with AUTOSAR version mixed is used
Description:	In previous releases of EB tresos Studio, the service component generation in the Rte Editor failed with an error message if you used a project with AUTOSAR version <code>mixed</code> and AUTOSAR 3.x modules. This is fixed now.
ID:	TRESOS-3175

Summary:	System Description Importer no longer removes existing SwcToEcuMappings and RootSwCompositionPrototypes
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Description:	When importing system description files into AUTOSAR 3.2 or higher system models, EB tresos Studio falsely removed existing SwcToEcuMappings and RootSwCompositionPrototypes if you enabled certain options. This bug is fixed now.
ID:	TRESOS-3213

Summary:	When reading or writing an AUTOSAR 3.2 parameter definition file containing a parameter with the config class PUBLISHED-INFORMATION, EB tresos Studio reported an error with code 1649
Description:	When you tried to load an AUTOSAR 3.2 parameter definition file via commandline or export such e file, EB tresos Studio falsely reported the error (1649) Autosar 2.1 features are only allowed for content type asc:2.1: PUBLISHED-INFORMATION. This bug is fixed now.
ID:	TRESOS-3233

Summary:	Commandline now handles Autosar 3.2 System Description correctly as input files for legacy commands.
Description:	Now the EB tresos Studio commandline does not show an error anymore when using Autosar 3.2 System Description files as input for ▶ legacy make ▶ legacy generate/verify ▶ legacy tree ▶ legacy vsmd
ID:	TRESOS-3197

Summary:	AUTOSAR 3.x system description files are verified immediately during import
Description:	From EB tresos Studio 12.0 onwards, AUTOSAR 3.x system description files have not been verified immediately during import. Therefore, the internal stored data might have been incomplete if the imported files would not contain required elements. As a result, editors and generators that rely on this data might not work correctly. For example, the behavior of the Rte Contract Phase Generator was undefined in this case. This problem has been fixed.
ID:	TRESOS-3178

Summary:	ECU-local system descriptions are now handled correctly
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Description:	In previous releases of EB tresos Studio, ECU-local system descriptions (also known as <i>system extracts</i>) were not handled correctly in AUTOSAR 4.0 projects. All delegation connector chains leading to the outside of the top level composition were ignored. This problem is fixed now.
ID:	TRESOS-3137

Summary:	The performance of verifying the ECU configuration has been improved
Description:	Due to an error in its caching mechanism, the verification of the ECU configuration wasted a lot of memory and took recognizable more time than actually needed. Since this effect grew exponentially with the size of the project, this problem especially came into account when e.g. huge system description files have been imported. This problem has been fixed.
ID:	TRESOS-3206

Summary:	The DBC Importer now supports variable value tables
Description:	In previous releases, the DBC Importer issued an error when parsing DBC files which contained environment variable value tables. The DBC Importer is now able to parse such files without issuing an error.
ID:	TRESOSWP-2176

Summary:	The Fibex Importer creates dedicated <code>ISignals</code> for each PDU
Description:	In previous releases, the Fibex Importer for Fibex 3.x files used to create for each signal in the Fibex file exactly one <code>ISignal</code> object in the system model. This <code>ISignal</code> object in the system model then used to be referenced by all <code>ISignalPDUs</code> that contained the signal. As required by AUTOSAR, the Fibex Importer now creates one dedicated <code>ISignal</code> object for each <code>ISignalPDU</code> that contains the corresponding signal.
ID:	TRESOS-3156

Summary:	<code>DataTypeMappingSets</code> are now set up correctly during the AUTOSAR 3.x to 4.0 model upgrade
Description:	Prior to this release, <code>DataTypeMappingSets</code> were set up incorrectly during the AUTOSAR 3.x to 4.0 model upgrade: Not all required <code>DataTypeMappingSet</code> elements were actually created. Moreover, not all required references from <code>SwcInternalBehavior</code> elements to these <code>DataTypeMappingSets</code> were created. <code>DataTypeMappingSets</code> and the references from <code>SwcInternalBehavior</code> elements are now set up correctly.
ID:	TRESOS-3200

Summary:	Referrable elements in AUTOSAR 4.0 models are handled correctly now
Description:	In former versions of EB tresos Studio, entities that are derived from <code>Referrable</code> but not from <code>Identifiable</code> were not handled correctly. For those entities, the <code>shortName</code> at-

	tribute was not set by the System Description Importer and they could not be referenced using their AUTOSAR path. This problem is fixed now.
ID:	TRESOS-3158

Summary:	Fixed DBC importer handling of enumeration attributes
Description:	In previous releases the DBC importer processed enumeration attributes, i.e. attributes restricting their values to one within a predefined value list, in a special way: When the DBC importer encountered ("") as an enumeration attribute value, it issued a warning and replaced this value by the first value contained in the attribute's predefined value list. This special handling of "" has been removed. When the DBC importer encounters an enumeration attribute value that is contained in the enumeration's predefined value list, it imports this value. When the attribute value is not contained, the DBC importer reports a warning and does not import the value.
ID:	TRESOS-3173

Summary:	Editors and dialogs do not access wrong data anymore
Description:	In former versions of EB tresos Studio, wizards which did not use transactions for accessing the data model could read the wrong values if they had written data to the model before. This is now fixed. Please note: the effect caused depended on the implementation of each wizard. The wizards delivered as part of the EB tresos Autocore were not affected.
ID:	TRESOS-3192

Summary:	Optional choice containers are no longer disabled after import
Description:	In former versions of EB tresos Studio, if you imported an AUTOSAR 4.0.x ECU configuration file choice containers stayed disabled. This happened if the module contained an optional choice which is disabled by default. The inner values, however had been imported correctly. This is now fixed, and the choice container is enabled as expected.
ID:	TRESOS-3236

Summary:	Existing service port mappings stay valid when using the System Description Importer
Description:	In earlier versions of EB tresos Studio, existing Service Port mappings were invalidated under some conditions when a System Description Importer was run. The problem occurred, if the importer option <i>Enable system model import</i> was selected and the option

	<i>Add software component ECU extract was not set to Do no import software components and mappings.</i> This issue has been resolved, existing Service Port mappings are preserved during import.
ID:	TRESOS-3165

1.10.2. Changes/new features

Summary:	EB tresos Studio supports the Ethernet and TCP/IP parts of AUTOSAR 4.1.0
Description:	EB tresos Studio provides support for the Ethernet and TCP/IP relevant parts of the system model as of AUTOSAR 4.1.0 (snapshot from 2012-08-09, XSD document version 4.-3.0, revision 0). You may now import and export AUTOSAR 4.1.0 system description files.
ID:	TRESOSWP-2219

Summary:	EB tresos Studio provides support for Measurement and Calibration data
Description:	EB tresos Studio now supports the AUTOSAR model parts required for Measurement and Calibration as defined in the AUTOSAR release 4.0.3. You can now import and export AUTOSAR system description files containing those entities.
ID:	TRESOSWP-2204

Summary:	EcuExtract is now created with a FlatMap
Description:	The EcuExtractCreator now additionally creates a FlatMap where all corresponding data prototypes, port prototypes and software component prototypes from the EcuExtract and the hierarchical system model are mapped together.
ID:	TRESOSWP-2214

Summary:	The system model upgrade from AUTOSAR 3.1/3.2 to 4.0 now stores platform data types in different packages
Description:	According to http://www.autosar.org/bugzilla/show_bug.cgi?id=52928#c11 , the AUTOSAR 3.1/3.2 to 4.0 system model upgrade stores the AUTOSAR 4.0 platform data types and their related entities in /AUTOSAR_Platform/ImplementationDataTypes, /AUTOSAR_Platform/BaseTypes and /AUTOSAR_Platform/DataConstrs. In previous releases, these entities were stored in /AUTOSAR/Platform/ImplementationDataTypes, /AUTOSAR/Platform/BaseTypes and /AUTOSAR/Platform/DataConstrs.
ID:	TRESOSWP-2153

Summary:	Sub-packages are now allowed for module definitions
Description:	In previous versions it has not been possible to define modules within any sub-package structure. This is required by AUTOSAR, when a vendor defines multiple complex device driver modules. You may now define modules within any sub-package structure. For backwards-compatibility reasons, the usage of sub-packages above module configurations is still not allowed.
ID:	TRESOSWP-2115

Summary:	System description importer can now import files with different meta model versions
Description:	Now it is possible to import files with different meta model versions. A new feature has been introduced to autodetect the meta model version of each file. For further information please visit the new and noteworthy document.
ID:	TRESOSWP-2207

Summary:	EB tresos Studio now supports AUTOSAR 3.2.2
Description:	EB tresos Studio now supports AUTOSAR 3.2.2. This includes: ▶ You may now import of AUTOSAR 3.2.2 system descriptions as ARXML files (new content type <code>sysd:3.2.2</code>). ▶ You may now import and export AUTOSAR 3.2.2 ECU configuration data from/into ARXML files (content type <code>asc:3.2</code>). ▶ The AUTOSAR 3.2.2 standard module definitions are installed together with EB tresos Studio within the sub directory <code>autosar\3.2</code>.
ID:	TRESOSWP-2159

1.11. Changes in version 13.0.0

1.11.1. Fixed problems

Summary:	The <i>TresosDB Importer</i> does not cause GUI freezes anymore
Description:	When you had created a <i>TresosDB Importer</i> in EB tresos Studio version 12.1, the GUI became unresponsive (i.e. frozen for some 10 seconds) after you had set the name of the file to <i>import</i> . This problem is fixed now.
ID:	TRESOS-3121

Summary:	The Generic Configuration Editor now always shows the menu item <i>Go to referenced node</i>
Description:	In EB tresos Studio version 12.1, the Generic Configuration Editor did not provide the menu item <i>Go to referenced node</i> when the module had used a reference expression of the path-type <code>XPathDataOfSchema</code> . This problem is fixed now.
ID:	TRESOS-3119

Summary:	AUTOSAR 3.1 to 4.0 system model upgrade now correctly upgrades system models with transport layer configuration references
Description:	In EB tresos Studio release 12.1, when you had upgraded an imported system model from AUTOSAR 3.1 to 4.0 that contained two or more communication clusters containing transport layer data, this resulted in two or more transport layer configuration elements in the 4.0 system model. The reference from the system element to these configuration elements, however, was set incorrectly. This defect has been resolved.
ID:	TRESOS-3090

Summary:	AUTOSAR 3.1 to 4.0 system model upgrade now correctly upgrades initial values of inter-runnable variables
Description:	In EB tresos Studio release 12.1, when you had upgraded an imported system model from AUTOSAR 3.1 to 4.0 that contained inter-runnable variables owning initial values, those values were not converted correctly. This defect has been resolved.
ID:	TRESOS-3101

Summary:	The DBC Importer now processes DAD files
Description:	Prior to fixing this problem, the DBC Importer had ignored DAD files that had been added to an importer configuration. This defect has been removed.
ID:	TRESOS-3095

Summary:	The tresosDB-API does not allow to reference from a persistent to a non-persistent entity anymore
Description:	The tresosDB-API allowed to set a reference from a persistent entity to a non-persistent entity. When saving the persistent entity, the reference pointing to the non-persistent entity was then silently ignored. This problem has been fixed: The API now throws an exception.
ID:	TRESOS-3092

Summary:	Mixing the configuration variants <code>VariantPostBuild</code> and <code>VariantPostBuild-Loadable</code> within one module definition does not cause a VSMD check failure anymore
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Description:	The vsmd check falsely reported a violation of the rule <code>[ecuc_sws_6051]</code> , when you had mixed the configuration variant <code>VariantPostBuild</code> and <code>VariantPostBuild-Loadable</code> within one module definition. This problem has been fixed.
ID:	TRESOS-3094

Summary:	Reuse of file locations in the RCtxt API do not lead to unexpected results anymore
Description:	Prior to fixing this problem, when an <code>RCtxt</code> object had been created several times for the same file location, EB tresos Studio would not dispose the resource. In other words, new <code>RCtxt</code> objects were not empty, when created for a location pointing to a file path that had been used before. This problem has been fixed.
ID:	TRESOS-2929

Summary:	The RCtxt API now considers the correct content type
Description:	Prior to fixing this problem, when a resource with the same file name had been created several times in a row with a different content type, EB tresos Studio would not set the new content type correctly, but would export the file as XDM instead. This problem has been fixed.
ID:	TRESOS-2930

Summary:	The options <code>-u</code> and <code>-a</code> do not swallowed the next item on the EB tresos Studio command line anymore
Description:	The EB tresos Studio command line offers the options <code>-u</code> (“Use Unix line endings”) and <code>-a</code> (“Verify all modules, also those that are not generated”) to be used with several commands. Those options are toggle options, i.e. they require no further parameters. In previous EB tresos Studio versions, the option <code>-u</code> when used with the commands <code>legacy convert</code> or <code>legacy vsmd</code> , and the option <code>-a</code> when used with the command <code>generate</code> had been parsed as parameter option. Therefore, the next command line parameter beneath these option was swallowed. This problem has been fixed.
ID:	TRESOS-2846

Summary:	The performance problems when saving projects and generating code has been fixed
Description:	In the previous version of EB tresos Studio, saving the project or generating code took too long whenever the project contained many files. This problem is fixed now.
ID:	TRESOS-3075

Summary:	Fibex 3.x Importer now creates correct E2E entities, if E2E PDUs are referenced by PDU-triggerings of different channels
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Description:	In previous version of EB tresos Studio, the Fibex 3.x Importer created faulty E2E entities when E2E PDUs had been referenced by <code>PDU-TRIGGERINGS</code> of different <code>CHANNELS</code> . This resulted in E2E signal groups that were not received as defined in the Fibex file. This problem is fixed now.
ID:	TRESOS-3044

Summary:	AUTOSAR 3.1 to 4.0 system model upgrade converts <code>ParameterRequireComSpec</code> and <code>ParameterProvideComSpec</code> elements correctly now
Description:	In EB tresos Studio version 12.1, when you had upgraded an imported system model from AUTOSAR 3.1 to 4.0 that contained <code>ParameterRequireComSpec</code> s or <code>ParameterProvideComSpec</code> s the model upgrader issued an error message <code>Failed to upgrade database: Error in insertion statement: Code: 0 Message: SWC2_RPortComSpec.RPortComSpec_Type may not be NULL in _File: commandfiles/swcd/20_SWCompPortsInterfacesDetails.sql...</code> This problem is fixed now.
ID:	TRESOS-3041

Summary:	The <code>DEST</code> attribute in ECU configuration files now contains the correct value
Description:	In previous versions of EB tresos Studio, if you used EB tresos Studio to create AUTOSAR 4.0.2 and 4.0.3 ECU configuration files contained a wrong value for the <code>DEST</code> attribute when these ECU configuration files contained a foreign reference, e.g. <code>ECUC-REFERENCE-VALUE</code> . The value of the <code>DEST</code> attribute in the <code>VALUE-REF</code> XML tag would then be always <code>IDENTIFIABLE</code> . This problem is fixed now.
ID:	TRESOS-3051

Summary:	ECU configuration files are not written with mixed line endings anymore
Description:	In previous version of EB tresos Studio, ECU configuration files could have been written with a mix of Unix line endings (<code>\n</code>) and Windows line endings (<code>\r\n</code>) when the configuration option <i>Use Unix line endings for configuration data files</i> had been disabled. This problem is fixed now.
ID:	TRESOS-2837

Summary:	The label of the continuous verification setting has been improved
Description:	In previous versions of EB tresos Studio, the preference setting for <i>continuous</i> verification was labeled Enable verifier. This label was ambiguous as it could have been misunderstood as an option for disabling the verification as a whole. This misunderstanding was aggravated by the fact, that the setting Verify readonly nodes could not be changed, when the Enable verifier setting was disabled. Now the continuous verification setting is labeled Enable continuous verification, and you can always change the Verify readonly nodes setting.

ID:	TRESOS-2588
Summary:	The misleading preference setting Automatically close projects that cannot be loaded has been removed
Description:	In previous versions of EB tresos Studio, when the user had tried to open a project with a corrupt <code>.prefs/preferences.xdm</code> file, the project would either be left open or closed afterwards, depending on the preference setting Automatically close projects that cannot be loaded. This preference was misleading as it could also be understood in a way that projects with invalid module configuration files would be closed when they could not have been loaded. The preference is now removed and the project will be in an open state, in which the users have a chance to manually fix the corrupt <code>.prefs/preferences.xdm</code> file, and then close and reopen the project themselves.
ID:	TRESOS-3001
Summary:	Loading and reloading a project with an error in the configuration file now both prompt a confirmation dialog
Description:	In previous versions, EB tresos Studio showed a different behavior for loading or reloading a project with an error in a configuration file. During the reload of a project, EB tresos Studio always removed the faulty module, while loading such a project opened a confirmation dialog. The behavior is now unified, EB tresos Studio now also prompts the user with a confirmation dialog when reloading a faulty module configuration.
ID:	TRESOS-3012
Summary:	Loading a project does not result in unsaved module configurations anymore
Description:	In previous versions of EB tresos Studio, when you had loaded a newly created project, some module configurations might have falsely shown unsaved changes. This problem has been fixed.
ID:	TRESOS-3036
Summary:	The vendor-specific module definition (VSMD) checker no longer throws a null pointer exception for the content type <code>asc:4.0.3</code>
Description:	When using the vendor-specific module definition (VSMD) checker of EB tresos Studio version 12.1.0, a null pointer exception could have been thrown. This error only occurred, when you had used the rules for AUTOSAR version 4.0.3 and the vendor-specific module definition (VSMD) had been missing a mandatory element. This problem has been fixed, the appropriate rule violation is reported now.

ID:	TRESOS-3039
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Summary:	The default selection of system models has been changed for the AUTOSAR release version mixed in the <i>New Project Wizard</i>
Description:	In previous versions of EB tresos Studio, EB tresos Studio had reported warning or error messages about not existing AUTOSAR elements even though the elements did exist in the input data. These error messages could occur in all parts of EB tresos Studio, and even in other EB tresos products based on EB tresos Studio, from which the system model was accessed. The error message occurred when you had selected <i>Autosar Release Version: mixed</i> in the <i>New Project Wizard</i>. EB tresos Studio would then select both system models for AUTOSAR version 3.x and AUTOSAR version 4.x by default. If you had then imported DBC, Fibex or AUTOSAR system description files, the content of the files might have ended up in the wrong system model instance. Products that use the EB tresos Studio APIs to access the system model might not have been able to deal with both instances. In this case, the error message was shown. To solve this problem, the default selection for projects with <i>Autosar Release Version: mixed</i> has been changed. If you now choose <i>Autosar Release Version: mixed</i>, only the AUTOSAR 4.x system model is chosen by default. However, you may change this default setting so that other configuration files are selected, too. Additionally, warnings are now displayed when you import different system model versions that may lead to problems with third-party products.
ID:	TRESOS-3060

Summary:	You may now import files with a relative path with the <i>Autosar Importer</i>
Description:	Prior to fixing this problem, when you had run an <i>Autosar Importer</i> with a relative path to the imported file, an error message would occur informing that relative paths cannot be resolved. This problem has been fixed: Now the relative path is converted into an absolute path to the working directory.
ID:	TRESOS-3053

Summary:	EB tresos Studio now validates still uncommitted user input when closing guided configuration editors
Description:	Prior to fixing this problem, when you had changed values in a guided configuration editor without committing them (e.g. by pressing <code><keycab>Return</keycab></code> or by selecting another GUI widget) and you had then directly closed the guided configuration editor, the change was not validated. That way EB tresos Studio may have run the guided configuration editor on an erroneous configuration. This problem has been fixed.

ID:	TRESOS-3116
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1.11.2. Changes/new features

Summary:	The Fibex 3.x Importer can derive a communication controller's bus type from the connected cluster
Description:	The Fibex 3.x Importer now also imports communication controllers which do not contain any information concerning the bus type the controllers support, e.g. CAN or FlexRay. If no such information is available, the Fibex 3.x Importer derives the bus type from the <code>PROTOCOL</code> parameter of the cluster the communication controller is connected to.
ID:	TRESOSWP-2122

Summary:	EB tresos Studio now supports AUTOSAR 3.2
Description:	EB tresos Studio now supports AUTOSAR 3.2. The support for AUTOSAR revision 3.2.1 in EB tresos Studio includes: ▶ You may now import of AUTOSAR 3.2.1 system descriptions as ARXML files (new content type <code>sysd:3.2.1</code>). ▶ You may now import and export AUTOSAR 3.2.1 ECU configuration data from/into ARXML files (new content type <code>asc:3.2</code>). ▶ EB tresos Studio now provides a new VSMD rule set named <code>asc:3.2</code> for validating vendor-specific module definitions against the AUTOSAR 3.2.1 standard. ▶ The maximum <code>SHORT-NAME</code> length for AUTOSAR 3.2 projects is restricted to 128 characters, which conforms to the AUTOSAR specification. However, for earlier AUTOSAR versions, you may still disable the length check. ▶ The AUTOSAR 3.2.1 standard module definitions are installed together with EB tresos Studio within the sub directory <code>autosar\3.2</code>.
ID:	TRESOSWP-1900

Summary:	AUTOSAR 3.1 to 4.0 system model upgrade now allows the upgrade of frames transmitted on different bus systems
Description:	In EB tresos Studio release 12.1, when you had upgraded an imported system model from AUTOSAR 3.1 to 4.0 that contained one or more frames that were sent over different bus systems, e.g. CAN and FlexRay, an error message was issued indicating that an upgrade is not possible due to these frames. In EB tresos Studio 13.0, the system model upgrade has been extended to support the upgrade of AUTOSAR 3.1 system models that contain such frames.
ID:	TRESOSWP-2084

Summary:	The ECU extract creation for AUTOSAR 4.0 system models now handles signal mappings
Description:	The mechanism for creating the ECU extract now determines all signal mappings that are relevant for the delegation port of the ECU extract composition and adds these signal mappings to the system of the ECU extract.
ID:	TRESOSWP-1857

Summary:	The C-data-structures-generator now supports separate generation of the identifier and the initializer for the definition of variables
Description:	The C-data-structures-generator now supports separate generation of the identifier and the initializer for the definition of variables. To generate just the identifier, you can use the method <code>CBackend.typedIdentifier()</code> . To generate the initializer, you can use the method <code>CBackend.initializer()</code> . For more information about these methods, refer to the Public API JavaDoc description of the class <code>CBackend</code> .
ID:	TRESOSWP-2123

Summary:	EB tresos Studio now supports configuration classes
Description:	EB tresos Studio now supports configuration classes. This has the following effects: ▶ In the <i>New Project Wizard</i> and in the <i>Properties for <project_name></i> dialog you can now change the configuration time setting of the project. ▶ The Generic Configuration Editor GUI now disables parameters of not editable configuration classes. ▶ In the VSMD checks a new EB tresos Studio-specific rule checks the attribute <code>post-BuildChangeable</code> according to [ecuc_sws_2006]. ▶ The <i>AUTOSAR Importer and Exporter</i> allows a limitation to specific configuration classes. ▶ The code generators take the configuration time and the configuration class of parameters into account. ▶ All API functionalities that access parameter values take the configuration time and the configuration class of parameters into account. The API functionalities can also query the configuration class and the configuration time of parameters. For detailed information on the configuration class support, refer to the EB tresos Studio documentation.
ID:	TRESOSWP-1899

Summary:	User documentation XPath API examples and description have been revised
Description:	The description and example of the XPath API functions have been revised. Incorrect examples have been replaced with examples that match the implementation.

ID:	TRESOSWP-2132
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1.12. Changes in version 12.1.0

1.12.1. Fixed problems

Summary:	The XPath bit functions now check for allowed bit values
Description:	The XPath functions <code>bit:bitset</code> , <code>bit:bitclear</code> , and <code>bit:getbit</code> now check whether the provided bit number is in the range <code>[0, 55]</code> . If the bit number is not in this range, the functions now produce an error message.
ID:	TRESOS-3096

Summary:	Opening <code>preferences.xdm</code> project files in read-only mode is now possible
Description:	In previous versions of EB tresos Studio, the attempt to open a project with <code>preferences.xdm</code> set to a <code>readonly</code> failed. In this case the project was closed again and an error was reported. This behavior has been changed. Now, the project opens up and the configuration can be loaded. However, an error is reported to you nonetheless, as all changes to the preferences cannot be saved.
ID:	TRESOS-3002

Summary:	References pointing from the EcuC configuration to the system model now work correctly
Description:	Prior to fixing this problem, <i>foreign references</i> - as defined by AUTOSAR - that point from the EcuC configuration into the system model did not work and therefore caused a warning message to open up in the Generic Configuration Editor. This problem has been fixed.
ID:	TRESOS-2834

Summary:	Default values of memento attributes are no longer overwritten
Description:	In previous releases of EB tresos Studio, when you had subclassed <code>AbstractMementoFactory</code> and had initialize attributes of mementos with default values inside the <code>getMemento</code> method and if the memento was used inside a table or tree-table, these values would have been overwritten by the column description default values.

	This problem has been fixed. The column description default values are now only used, if the memento attribute is not set.
ID:	TRESOS-2963

Summary:	You can now register guided configuration triggers and push services only with the major version number of an AUTOSAR release
Description:	In the previous version of EB tresos Studio, it was not possible to register guided configuration triggers or push services for one specific AUTOSAR major version number without also specifying the minor version number. This problem has been fixed.
ID:	TRESOS-3008

Summary:	The Generic Configuration Editor is now in sync with the <i>Outline</i> view
Description:	In the previous version of EB tresos Studio, under certain circumstances the selected node in the Generic Configuration Editor was not in sync with the selected node in the <i>Outline</i> view. This problem has been fixed.
ID:	TRESOS-2923

Summary:	Generic Configuration Editor now disposes GUI handles correctly
Description:	In the previous version of EB tresos Studio, the Generic Configuration Editor did not dispose user interface handles in the Calculate button. This led to the error message <code>org.eclipse.swt.SWTError: No more handles</code> that opened up after you have worked with EB tresos Studio for some time. This problem has been fixed
ID:	TRESOS-2979

Summary:	NaN, INF and -INF are now allowed values for the MIN and MAX attribute of float parameter definitions
Description:	According to the AUTOSAR specification, the values NaN, INF and -INF are allowed values for the float parameter attributes MIN, MAX or DEFAULT. In previous EB tresos Studio releases, an error was reported when a file containing these values was loaded into EB tresos Studio. This problem has been fixed. You may now load an AUTOSAR 2.x and 3.x file containing NaN, INF and -INF as values for the parameter attributes MIN, MAX or DEFAULT into EB tresos Studio without errors.
ID:	TRESOS-2333

Summary:	Files are now generated for custom generation modes
Description:	In the previous version of EB tresos Studio, when you had started code generation for a custom generation mode such as <code>generate_swcd</code> and the <i>Project Explorer</i> view was currently not active, no files were generated. This is fixed now.
ID:	TRESOS-2954

Summary:	Now, the <code>ORIGIN</code> attribute is written correctly for <code>ECUC-ENUMERATION-LITERAL-DEFS</code>
Description:	In the previous version of EB tresos Studio, the <code>ORIGIN</code> attribute for <code>ECUC-ENUMERATION-LITERAL-DEF</code> was always set to the value <code>unknown</code> . Now, the correct value is written for the <code>ORIGIN</code> attribute.
ID:	TRESOS-2848

Summary:	Corrupt projects can now be deleted directly after creation
Description:	In the previous version of EB tresos Studio, when you had accidentally created a corrupt project e.g. by using a module with an invalid configuration schema XDM file, the project could not be deleted without restarting EB tresos Studio. This is fixed now.
ID:	TRESOS-2854

Summary:	Disabling or enabling modules from the <i>Project Explorer</i> view is now possible
Description:	In previous versions of EB tresos Studio, when you had disabled or enabled a module configuration using the <i>Project Explorer</i> context menu, this change has not been saved when you closed the project and opened it again. This problem is fixed now.
ID:	TRESOS-2997

Summary:	When you open an importer/exporter from the command line, the importer's/exporter's configuration is now validated
Description:	Previous versions of EB tresos Studio did not validate the configuration of an importer or exporter that was launched from the command line with the commands <code>tresos_cmd.bat import export <project> <importer exporter name></code>. This could lead to unexpected behavior of the importer/exporter. Now, the configuration of every importer/exporter is validated before it is launched via the command line. If an error is found, it is reported to you and the execution is aborted.
ID:	TRESOS-2710

Summary:	The <code>INFO</code> message about system description files being read is not printed multiple times on the command line anymore
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Description:	When working with AUTOSAR system description files on the command line, the <code>INFO</code> message that informed that the system description file is read, was printed multiple times under some circumstances. This error is fixed now.
ID:	TRESOS-2827

Summary:	Studio does not swallow command line options in legacy mode anymore
Description:	Prior to fixing this problem, when you had called EB tresos Studio in command line mode, several command line options were swallowed and therefore their default was used. As one possible consequence, the given configuration files were loaded in a wrong way and therefore generation may have led to a different result than in previous EB tresos Studio releases. Affected of this problem were the commands legacy convert, legacy generate, legacy make and legacy verify if they were equipped with at least one the options overwriteAttributes, mergeAttributes, mergeConfigs, verifyReadOnly or suppress-ReadOnlyWarnings. This problem has been fixed.
ID:	TRESOS-2998

Summary:	Calculation of default configuration parameters do not use outdated input anymore
Description:	Prior to fixing this problem, when a calculated default value had referred to another field on the same page of the Generic Configuration Editor, the last user input was ignored if you had clicked the Calculate button immediately after changing the value in the referred input field without pressing Enter. This problem has been fixed.
ID:	TRESOS-2921

Summary:	AUTOSAR ECU Configuration Exporter does not write disabled parameters anymore
Description:	Prior to fixing this problem, the AUTOSAR ECU Configuration Exporter wrote parameters of the AUTOSAR module ECU Configuration even if they had been disabled. This problem concerned parameters that were disabled via an XPath expression in the module at the associated parameter definition. The EPC Exporter was not affected by this problem. This problem is fixed now.
ID:	TRESOS-2940

Summary:	The ContentType <code>asc:4.0</code> has been renamed to <code>asc:4.0.2</code>
Description:	The ContentType <code>asc:4.0</code> has been renamed to <code>asc:4.0.2</code>. This was necessary, because ever since the introduction of the ContentType <code>asc:4.0.3</code>, AUTOSAR did not change the namespace between these two versions but introduced incompatible changes to the meta model.

	To distinguish between version 4.0.2 and 4.0.3 in exported ARXML files, the XML attribute <code>xsi:schemaLocation</code> is used. When exporting files in the AUTOSAR format, EB tresos Studio sets the <code>xsi:schemaLocation</code> to <code>http://autosar.org/schema/r4.0 AUTOSAR_4-0-2.xsd</code> for an AUTOSAR 4.0.2 file and to <code>http://autosar.org/schema/r4.0 AUTOSAR_4-0-3.xsd</code> for an AUTOSAR 4.0.3 file. If you want to read an AUTOSAR file with the namespace <code>http://autosar.org/schema/r4.0</code> that does not contain one of the strings 4-0-2 or 4-0-3 in the <code>xsi:schemaLocation</code>, you have to specify the correct <code>ContentType</code> manually. Otherwise an error will occur. The EB tresos Studio command line no longer supports the <code>ContentType asc:4.0</code>. Instead you have to specify either <code>asc:4.0.2</code> or <code>asc:4.0.3</code>. Existing configurations of the AUTOSAR Importer and Exporter automatically switch from <code>ContentType asc:4.0</code> to <code>asc:4.0.2</code>.
ID:	TRESOS-2945
Summary:	The attribute <code>ConditionalChangeNad.id</code> in the AUTOSAR 4.0.2 system template is handled correctly now
Description:	The attribute <code>ConditionalChangeNad.id</code> in the AUTOSAR 4.0.2 system template was not correctly imported in former versions of EB tresos Studio. This error is fixed now. Note that this attribute will be reset to the default value when the model is loaded if it has been imported with an earlier version of EB tresos Studio.
ID:	TRESOS-2949
Summary:	The presentation of children of added/removed container nodes in the <i>Compare</i> editor for XDM files has been improved
Description:	When comparing XDM files in EB tresos Studio, the <i>Compare</i> editor for XDM files did not properly display nodes that had been added or removed as children of some added/removed containers, especially when the container had more children that fitted into the currently visible area of the editor. This problem is fixed now.
ID:	TRESOS-2922
Summary:	The <i>Sidebar</i> view is now up-to-date after loading the first project
Description:	Prior to fixing this problem, when you had loaded the first project after starting EB tresos Studio, the <i>Sidebar</i> view only showed contents after you selected the project in the <i>Project Explorer</i> view once again. This problem is fixed now.
ID:	TRESOS-2675
Summary:	References to existing elements in the AUTOSAR system model are not annotated with a warning anymore

Description:	Prior to fixing this problem, references to some types of entities in the AUTOSAR system model were always annotated with a warning marker although the referenced element did exist. For those references, no targets were offered in the drop-down list box either. This problem is fixed now.
ID:	TRESOS-2935

Summary:	The attribute <code>DEST</code> does not have a wrong value anymore when converting AUTOSAR 4.0 ECUC files
Description:	Prior to fixing this problem, when you used the <code>legacy convert</code> command to convert a configurations file to the AUTOSAR 4.0 file format, the <code>DEST</code> attribute was not written correctly for reference nodes. In these cases, the static value <code>IDENTIFIABLE</code> was used instead of the correct value. This problem is fixed now
ID:	TRESOS-2914

Summary:	A parameter value whose <code>DEFAULT</code> value is calculated with an erroneous calculation formula can now be set manually
Description:	Prior to fixing this problem, when a parameter did not have a value and its <code>DEFAULT</code> value had been calculated with a calculation formula that led to an error, the value of this parameter could not be set. In this case, you could not set the value of the parameter neither via public API nor in the GUI. This problem has been fixed.
ID:	TRESOS-2909

Summary:	Warnings do not appear anymore when loading a module with a Pre-configuration
Description:	Prior to fixing this problem, when you loaded a module with a <i>Pre-configuration</i> , two warnings appeared stating that the <i>EDITABLE</i> and the <i>READONLY</i> attribute would be removed. Now, these warnings do not appear anymore.
ID:	TRESOS-2908

Summary:	A <code>NullPointerException</code> does not occur in class <code>ModuleConfigContext</code> anymore when switching the focus back and forth in different views
Description:	Prior to fixing this problem, when you switched the focus back and forth between different views, a <code>NullPointerException</code> occurred in the class <code>ModuleConfigContext</code> . This problem has been fixed.
ID:	TRESOS-2895

Summary:	Renaming <code>SHORT-NAMES</code> of AUTOSAR 4.0 models using the public API is now possible
Description:	In EB tresos Studio release 12.0, attempts to change <code>SHORT-NAMES</code> of AUTOSAR 4.-0 data models by using the public API always resulted in the error code 3112 (<code>SHORT-NAME_CASESENSITIVE_CLASH_CANNOT_ADD</code>). This problem is fixed now.

ID:	TRESOS-2883
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Summary:	The <i>Problems View</i> does not report errors for disabled module configurations anymore
Description:	Prior to fixing this problem, the <i>Problems View</i> sometimes reported errors for disabled module configurations. This problem is fixed now.
ID:	TRESOS-2897

Summary:	When importing or loading a system description 3.x file, no <code>NullPointerException</code> is thrown anymore
Description:	Prior to fixing this problem, a <code>NullPointerException</code> might have been thrown during import or when loading such a file. This depended on the content of the AUTOSAR 3.x system description file. This problem has been fixed.
ID:	TRESOS-2847

Summary:	In the <i>Module Configuration</i> dialog, no false confirmation dialog appears anymore
Description:	In the <i>Module Configurations</i> dialog, when you disable a mandatory module configuration or when you disable the code generation for a mandatory module configuration, a confirmation dialog is shown. Prior to fixing this problem, this confirmation dialog was also shown if you had not selected a mandatory module configuration selected, or if the selected mandatory module configuration had already been disabled. This problem is fixed now.
ID:	TRESOS-2879

Summary:	The AUTOSAR 3.1 to AUTOSAR 4.0 system model upgrade has been improved
Description:	The AUTOSAR 4.0 parameters <code>NmEcu.nmComControlEnabled</code>, <code>NmEcu.nmMultipleChannelsEnabled</code>, <code>NmEcu.nodeDetectionEnabled</code>, <code>CanNmCluster.nmRepeatMessageTime</code>, <code>NmPdu.nmVoteInformation</code>, and <code>NmPdu.nmDataInformation</code> are now converted correctly from their counterparts in the AUTOSAR 3.1 system model. The AUTOSAR 4.0 entities <code>FlexrayNmClusterCoupling</code> and <code>CanNmClusterCoupling</code> are now also created for clusters that are not connected to other clusters. Thus, the parameters <code>FlexrayNmClusterCoupling.nmControlBitVectorEnabled</code>, <code>FlexrayNmClusterCoupling.nmDataDisabled</code>, <code>FlexrayNmClusterCoupling.nmScheduleVariant</code>, <code>CanNmClusterCoupling.nmBusloadReductionEnabled</code>, and <code>nmImmediateRestartEnabled</code> become available for these clusters. The parameters <code>CanNmCluster.nmCbvPosition</code>, and <code>CanNmCluster.nmNidPosition</code> are now converted correctly from bit offset in AUTOSAR 3.1 to byte offset in AUTOSAR 4.0.

	The <code>FlexrayCluster</code> custom parameter <code>FrNmRepeatMessageBitEnabled</code> in the AUTOSAR 3.1 system model is now converted to the <code>FlexrayNmCluster</code> custom parameter <code>NmRepeatMessageBitEnabled</code> in the AUTOSAR 4.0 system model.
ID:	TRESOS-2886

Summary:	In the <i>Project Explorer</i> view, no false confirmation dialog appears anymore
Description:	When you disable a mandatory module configuration via the context menu of the <i>Project Explorer</i> view, a confirmation dialog is shown. Prior to fixing this problem, this confirmation dialog was also shown, if the mandatory module configuration had already been disabled. This problem is fixed now.
ID:	TRESOS-2887

Summary:	The Fibex 3.x Importer does not create superfluous PDU triggering for Nm-PDUs and N-PDUs anymore
Description:	Prior to fixing this problem, the Fibex 3.x Importer created <code>PDUTriggering</code> elements for Nm-PDUs and N-PDUs, if the send/receive relationships of an <code>EcuInstance</code> was modeled using the <code>COMPLETE-FRAME</code> FIBEX XML tag. However, the AUTOSAR 3 system model does not define <code>PDUTriggering</code> elements for these PDU types. This problem has been solved.
ID:	TRESOSWP-1917

Summary:	The Fibex 3.x Importer does not ignore duration parts equal to or greater than one minute
Description:	In previous releases, the Fibex 3.x Importer configured durations by only taking the duration part into account that is less than one minute. Parts equal to or greater than one minute were implicitly ignored during the importer run. This problem has been solved.
ID:	TRESOS-2905

Summary:	The code generator now waits for the model verification to be finished
Description:	In previous versions of EB tresos Studio, when you started the code generator, it did not wait for a currently running model verification to be finished. This potentially caused two types of misbehavior: ▶ If there had been errors in your project that were already fixed, you were still asked whether code generation should be continued. ▶ In other cases, if there were still existing errors in your project that have not been fixed, you were not asked whether code generation should be continued. These problems are fixed now.
ID:	TRESOS-2899

Summary:	No stack overflow in XPath public API occurs anymore
Description:	In EB tresos Studio version 12.0.0, a stack overflow occurred if the XPath functionality of the <code>DCtxt</code> API was used too often (i.e. about 13.000 to 14.000 times). This problem is fixed now.
ID:	TRESOS-2892

Summary:	XDM files of removed module configurations are now deleted
Description:	In EB tresos Studio release 12.0, when you removed a module configuration with unsaved changes, the configuration XDM files were not deleted. This problem is fixed now.
ID:	TRESOS-2889

Summary:	The <i>Element Outline</i> views now show the selected element, if you open the Generic Configuration Editor from the <i>Problems View</i>
Description:	Prior to fixing this problem, when you opened the Generic Configuration Editor via the <i>Problems View</i> , the <i>Element Outline</i> views were showing the root-element of the editor and not the selected element. This problem is fixed now and the <i>Element Outline</i> views show the selected element.
ID:	TRESOS-2863

Summary:	AUTOSAR 3.1 to 4.0 system model upgrade converts mapping elements correctly now
Description:	In EB tresos Studio release 12.0, when you upgraded an imported system model from AUTOSAR 3.1 to 4.0, the contained <code>SwCompToImplMappings</code> were upgraded in such a way that they removed previously upgraded <code>SwcToEcuMappings</code> in the 4.0 model. This problem is fixed now.
ID:	TRESOS-2904

Summary:	New public API class informs about events cleared from the SWTTable
Description:	With the new public API class <code>dreisoft.tresos.guidedconfig.api.gui.event-ing.TableDeselectionEvent</code> you are now informed about events cleared from the SWTTable.
ID:	TRESOS-2737

Summary:	System description verification messages now appears in a more detailed dialog
Description:	Messages created during a system description verification are now presented as tree in which errors, warnings and infos are separated.
ID:	TRESOS-2924

Summary:	System description verification messages provide more information now
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Description:	Messages created during a system description verification now inform you about the nearest identifiable entity in which the verification error occurred.
ID:	TRESOS-2919

Summary:	The <i>Workflows</i> view now shows project-dependent actions even if a file resource is selected
Description:	In previous versions of EB tresos Studio, if you had selected a file resource e.g. the <code>. - project</code> file, in the <i>Project Explorer</i> view, the <i>Workflows</i> view would display project-dependent actions as not available. Now the actions are available and can be performed.
ID:	TRESOS-2999

Summary:	Undo/Redo support now works correctly when deleting multiple entries in guided configuration tables
Description:	In previous versions of EB tresos Studio, problems occurred with undoing/redoin some operations. In these cases, an error message opened up after multiple entries were deleted from a table in a guided configuration editor. This issue has been fixed.
ID:	TRESOS-2985

Summary:	The component generating the ECU extract of the software component description can now handle nested compositions without reporting a <code>ClassCastException</code>
Description:	Previous versions of EB tresos Studio reported a <code>ClassCastException</code> during the generation of the flat ECU extract model from a AUTOSAR 4 software component description with nested software compositions. The ECU extract is typically generated after a system description import. This issue has been fixed and the error is not reported anymore.
ID:	TRESOS-3014

1.12.2. Changes/new features

Summary:	EB tresos Studio now provides support for AUTOSAR 4.0.3
Description:	EB tresos Studio now provides support for AUTOSAR 4.0.3 (XSD document version 4.2.-1, revision 3). You may now create AUTOSAR 4.0.3 compliant modules, configure them within EB tresos Studio and import and export AUTOSAR 4.0.3 files.
ID:	TRESOSWP-1975

Summary:	The <i>Compare</i> editor for XDM files now displays warnings if nodes are ignored
Description:	In previous versions of EB tresos Studio, the <i>Compare</i> editor for XDM files silently ignored nodes that did not match the structure of the original file.

	Now it displays warnings for all ignored nodes.
ID:	TRESOSWP-1944

Summary:	Improved rendering of choices in the Generic Configuration Editor
Description:	The rendering of choice containers within in the Generic Configuration Editor has been revised for better usability and performance. For details, please refer to the <i>EB tresos Studio new and noteworthy</i> document.
ID:	TRESOSWP-1925

Summary:	Team collaboration: automatic reload
Description:	In previous versions of EB tresos Studio, you had to manually trigger a reload of a loaded project whenever one of the relevant files had changed, e.g. after applying an update from your version control system. Now EB tresos Studio detects changes on those files and triggers the reload of a loaded project automatically.
ID:	TRESOSWP-1747

Summary:	A new public API class is now available to invoke arbitrary code generators
Description:	With the new public API class <code>dreisoft.tresos.launcher2.api.project.-Project</code> you can now invoke any arbitrary code generators on demand.
ID:	TRESOSWP-1913

Summary:	Team collaboration: The <i>Compare</i> editor for XDM files now supports comparing attributes
Description:	In previous versions of EB tresos Studio, the <i>Compare</i> editor for XDM files did not consider differences in node attributes except for changes of the <code>ENABLE</code> attribute of optional nodes. Now the <i>Compare</i> editor for XDM files supports comparing and merging differences in arbitrary XDM attributes.
ID:	TRESOSWP-1844

Summary:	Registration of target/derivative-independent ECU resource properties is possible now
Description:	In previous versions of EB tresos Studio, you had to declare a specific target and derivative in the registration of ECU resource properties. Now it is possible to register ECU resource properties that apply for any target and derivative for a specific module.
ID:	TRESOSWP-1908

Summary:	The Fibex 3.x Importer now supports signals in NM-PDUs
Description:	The Fibex 3.x Importer has been extended to support the import of signals that are residing in NM-PDUs.

ID:	TRESOSWP-1939
Summary:	XPath expressions with automatic attributes with the type Range are now supported
Description:	You can now use automatic attributes with the type <code>Range</code> with variable schema nodes to restrict the allowed values. In previous versions of EB tresos Studio, only constant values were allowed for comparison. Now you can also use XPath expressions that are evaluated at runtime.
ID:	TRESOSWP-1958
Summary:	The Com Importer now supports the configuration of arbitrary modules
Description:	Previous versions of EB tresos Studio only supported the configuration of a restricted set of AUTOSAR module types. Especially the configuration of OEM-specific modules was not possible. This restriction has been lifted.
ID:	TRESOSWP-1982
Summary:	DBC Importer support for signal groups and signal data types
Description:	The DBC Importer now supports the import of signal groups using the <code>SIG_GROUP_</code> token. Moreover, the DBC Importer now also supports the specification of signed integral data types as well as IEEE single and double precision data types for signals contained in the imported DBC file via the <code>SIG_VALTYPE_</code> token.
ID:	TRESOSWP-1946
Summary:	A new context menu for references is available
Description:	The new context menu <i>List referenceable nodes</i> has been introduced for references in the Generic Configuration Editor. If you select this new context menu, all referenceable nodes are listed in the <i>Search</i> view. If only one referenceable node exists, the command navigates directly to the node. This helps you to e.g. add new referenceable nodes to a list.
ID:	TRESOSWP-386
Summary:	Vendor-specific module definition (VSMD) rules for AUTOSAR 4.0.3 are available
Description:	New Vendor-specific module definition (VSMD) rules for AUTOSAR 4.0.3 have been introduced to check supported configuration variants, reference destinations and alphabetical ordering.
ID:	TRESOSWP-1987
Summary:	Adaptable undo history size for Guided configuration framework
Description:	You can undo/redo operations that have been done via the <code>MementoOperationHandler</code> . Now you can also adapt the undo history size via the <i>Preferences</i> dialog. To

	change the undo history, open the EB tresos Studio <i>Preferences</i> dialog, navigate to <i>General/Editors/Text Editors</i> and change the value for the Undo history size .
ID:	TRESOSWP-1992

Summary:	ECU extract for AUTOSAR 4.0.2 and 4.0.3 now supports port interface mappings
Description:	The creation of an ECU extract for AUTOSAR version 4.0.2 and 4.0.3 now handles the port interface mappings. The connectors of the resulting ECU extract contain the merged mappings.
ID:	TRESOSWP-1977

Summary:	ECU extract for AUTOSAR 4.0.2 and 4.0.3 now preserves service components and connectors between service ports
Description:	If you create an ECU extract for AUTOSAR version 4.0.2 and 4.0.3, all service components and connectors between service ports are now kept when you update the ECU extract, if the service components and connectors are not already contained in the newly created ECU extract.
ID:	TRESOSWP-1998